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A Survey of Wearable Devices and Challenges

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***Abstract*—As smartphone penetration saturates, we are witnessing a new trend in personal mobile devices—wearable mobile devices or simply *wearables* as it is often called. Wearables come in many different forms and flavors targeting different accessories and clothing that people wear. Although small in size, they are often expected to continuously sense, collect, and upload various physiological data to improve quality of life. These requirements put significant demand on improving communication security and reducing power consumption of the system, fueling new research in these areas. In this paper, we first provide a comprehensive survey and classification of commercially available wearables and research prototypes. We then examine the communication security issues facing the popular wearables followed by a survey of solutions studied in the literature. We also categorize and explain the techniques for improving the power efficiency of wearables. Next, we survey the research literature in wearable computing. We conclude with future directions in wearable market and research.**

***Index Terms*—Wearables, wearable sensors, wearable computers, smart watches.**

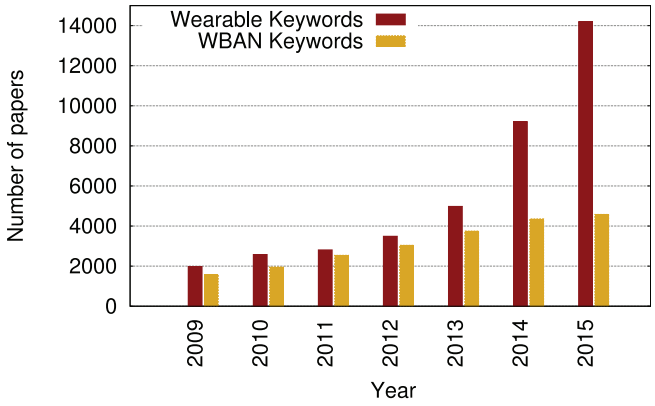


Fig. 1. Google Scholar hits for Wearable-related keywords (*"Wearable Devices", "Wearable Sensors", "Wearable Technology",* and *"Wearables"*) vs. WBAN-related keywords (*"Body Area Networks"* and *"Body Sensor Networks"*).

I. Introduction

WHILE the penetration rate for smartphones saturates, we are witnessing a fascinating new trend in the mobile device market. Users are showing an increasing interest in practically wearing a mobile device that promises to enhance the quality of life in a way that smartphones alone cannot achieve. These devices, which include smart watches, wrist bands, smart glasses, smart jewellery, electronic garments, skin patches, and so on, are often simply referred to as *wearables*.

Wearables can sense, collect, and upload physiological data in a 24x7 manner, providing opportunities to improve quality of life in a way not easily achievable with smartphones alone. Wearables can also help users perform many other useful micro tasks, such as checking incoming text messages and viewing urgent information, much more conveniently and naturally than possible with a smartphone, which is often carried in pockets or bags. Wearables are also being used to provide a range of value added services such as indoor localisation and navigation [1], [2], financial payments [3], [4], physical and mental health monitoring [5], [6], sport analytics [7], and medical insurance analytics [8]. According to recent market reports [9], [10], 44.4% increase in wearable shipments is forecasted for 2016, compared to the 80 million device shipments in 2015 and the yearly shipments is expected to reach 200 million in the year 2019. The wearable technology market is expected to reach a market value of $57,653 million by 2022, which is almost 3 times of that in 2016 ($19,633 million) [11].

The value proposition of wearables, especially the health monitoring aspects, has some overlap with another research topic known as wireless body area networks (WBANs) [12]. However, the vision of WBAN is much broader as it seeks to network together a large number[1] of wearable sensors and actuators, including medically implanted devices, to provide a very comprehensive and accurate monitoring of human health. Understandably, WBAN requires significant research efforts in physical, medium access control, and network layers before it can reach the commercial success. In contrast, wearables are low-cost, independent devices targeting personal use without any significant binding to medical professionals. This simplicity makes wearables more widely deployed, which in turn has led research attention at a much faster pace compared to WBAN (see Figure 1).

[1]WBAN targets interconnecting up to 50 devices [12].

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Over the last few years, a wide variety of wearables have appeared in the market offering different functionalities and wearing options. At the same time, a significant volume of scientific articles have been published in the literature highlighting various research challenges related to wearables. Although a few articles surveying the broader research topic of WBAN have been published in the past [12], they do not capture the recent developments related to wearables. In this article, we survey the current trends, technologies, research challenges and solutions for wearables. We also discuss the future market trends and research directions related to the design, development, and improvements of next generation wearables.

To the best of our knowledge this is the first survey about available wearable devices, prototypes, and overall research challenges. Several works looked into the very specific aspects of some technologies associated with wearables. For example, Hartman [13] provides an overview of available electronic components and toolkits to design and build smart garments. Similarly, Sazonov and Neuman [14] present the operating mechanisms of various sensors available for wearable devices and different fundamental algorithms associated with them such as step counting and activity recognition. Some work look into the use of wearable sensors in specific applications in the likes of *health monitoring and prognosis* [15]–[17], *activity recognition* [18], and *assisted living* [19]. While surveying *mobile crowd sensing* techniques, Zhang *et al.* [20] briefly surveyed the use of wearables as sensing units and similarly Harle [21] surveyed the use of wearables in *indoor positioning* applications. In contrast, in this survey we focus on consumer wearables, associated research challenges, available solutions, and potential future research directions.

The remainder of the paper is organised as follows. In Section II, we survey and classify the state-of-the-art wearables by examining the features and specifications of over 100 different products and research prototypes. Section III surveys the security vulnerabilities of the communication protocols used by commercial wearables and examines various approaches for addressing them. Section IV surveys research on improving the power consumption and battery life of wearables. Section V discusses research in wearable computing. Future directions and open research challenges are discussed in Section VI. Section VII concludes the paper.

II. State-of-the-Art of Wearable Devices

In this section, we first present a comprehensive survey of ***existing commercial wearable products*** under three categories: ***1) Accessories***, ***2) E-Textiles***, and ***3) E-Patches*** as illustrated in Figure 2. More specifically, we survey features, hardware specifications, and where applicable known limitations of over 100 commercial wearable products and provide an analysis on the communication technologies, battery technologies, battery life information, security features, and manufacturing material (if applicable).

Second, under ***research prototypes*** we survey research level prototypes that are either fitting into one of the three categories we defined above and provide novel features not available in commercial products, or new types of wearable devices that are yet to become mainstream. A summary of commercial products and research prototypes we surveyed is shown in Table I.

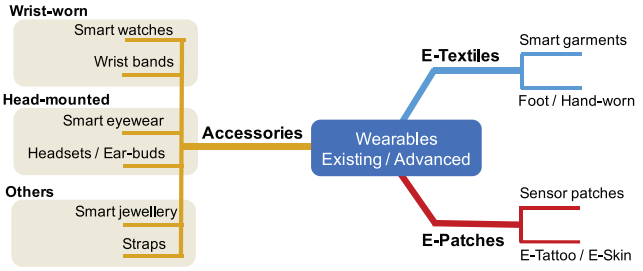


Fig. 2. Classification of wearable devices.

In this survey, we review only those devices that can be easily worn by the user without requiring any medical procedure. There are number of wearable devices such as ingestible sensors [22], wearable injectors [23]–[25], and wearable insulin pumps [26]–[28] that require some initial medical intervention to decide whether the user requires such device and to place it properly on the human body (i.e., devices that require Food and Drug Administration (FDA) approval [29]), are not considered under the scope of this survey.

A. Existing Products

1) Accessories: We define *Accessories* as the wearable devices worn externally by users and do not fit into the category of main clothing items. *Accessories* are by far the most popular category of devices in the current wearable market. Such devices include smart watches, wrist bands, smart glasses, chest straps, smart belts, and various clothing clip-ons. We next describe the existing products in Accessories under three sub categories along with their use cases.

a) Wrist-worns: Wrist-worns, as the name implies are the wearable devices that are worn on the wrist. The existing commercial wrist-worn products are either smart watches or wrist bands and we delve into details of these two types of products below.

Smart Watches: Smart watches are one of the most popular wearable device types. According to recent Gartner reports [10], in the wearables market smart watch sales are only second to Bluetooth headsets with approximately 50 million units predicted to be sold in 2016.

Usually the functionality of current smart watches are twofold. First, they act as ***communication and notification*** tools by complementing smartphones with features such as receiving notifications (e.g., phone calls, SMS, email, voice control, and weather updates) and perform micro interactions (e.g., launch an app in the phone, limited Web browsing, add reminders, and taking voice commands). Second, most of the smartwatches also can monitor some ***human physiological signals and biomechanics*** and thus act as fitness tracking devices that help users to log their daily activities such as automatically recording work out times, tracking heart rate, step counts, and calories burnt. The collected data is transferred to the smartphone or a cloud server for analytics and better presentation for users such as through dashboards.

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TABLE I
EXAMPLE PRODUCTS OF DIFFERENT WEARABLE CATEGORIES

	Description	Existing Products	Research Prototypes
Accessories			
Wrist-worns			
Smart watches	Wrist-worn devices with a touchscreen display.	Apple iWatch [30] Samsung Gear S2 [31] Moto 360 [32] Pebble Time [33]	*Smart-watch Life Saver [34]* *Finger-writing with Smart-watch [35]*
Wrist bands	Wrist-worn devices with fitness tracking capabilities or other functionalities, generally without a touchscreen display.	UP by Jawbone [36] Fitbit Flex [37] MOOV NOW [38] Nymi Band [39]	*Wrist-worn Bioimpedance Sensor [40]* *Wrist-worn Smoking Gesture Detector [41]* *Ultrasonic-speaker Embedded Wrist Piece and Neck Piece [42]*
Head-mounted Devices			
Smart eyewear	Spectacles or contact lenses with sensing, wireless communication, or other capabilities.	Microsoft HoloLens [43] FUNIKI Ambient Glasses [44] Recon Jet [45]	*Google Glass [46]* *Google Contact Lens [47]* *Object Modelling Eye-Wear [48]* *iShadow Mobile Gaze Tracker [49]* *Indoor Landmark Identification Supporting Wearables [50]* *Chroma [51]*
Headsets and Ear-buds	Bluetooth enabled headsets or ear plugs. Sensor-embedded hats and neck-worn devices are also found in research products.	Sony Xperia Ear [52] Apple AirPods [53] Bragi Dash Pro [54]	
Other Accessories			
Smart jewellery	Jewellery designed with features such as health-monitoring and handless-control.	Smarty Ring [55] Kerv [56] Bellabeat Leaf [57]	*Typingring [58]* *Gesture Detection Ring [59]*
Straps	Chest straps, belts, arm bands, or knee straps equipped with sensors for health tracking or other functionalities.	MYO Armband [60] Zephyr Bioharness [61]	*Pneumatic Armband [62]* *BodyBeat [63]*
E-Textiles			
Smart garments	Main clothing items that also serve as wearables such as shirts, pants, and undergarments.	Athos [64] Hug Shirt [65] Solar Shirt [66] Spinovo [67]	*Myovibe [68]* *Dopplesleep [69]*
Foot / Hand-worn	Shoes, socks, insoles, or gloves embedded with sensors.	Lechal [70] Sensoria [71] Fujitsu Gesture-control Gloves [72]	*LookUp [73]* *Gait Analysis Foot Worns [74]* *Foot-worn Inertial Sensors [75]*
E-Patch			
Sensor patches	Sensor patches that can be adhered to the skin for either fitness tracking or haptic applications.	HealthPatch MD [76] Thync [77] UPRIGHT [78]	*DuoSkin [79]* *Tattoo-Based Iontophoretic-Biosensing System [80]* *Smart Tooth Patch [81]*
E-Tattoo / E-Skin	Tattoos with flexible and stretchable electronic circuit to realise sensing and wireless data transmission.	Motorola e-tattoo [82] Wearable Interactive Stamp Platform [83]	

Moto 360 [32] is an example smart watch product. It runs the Android Wear operating system and works in complement to Android phones (limited set of functionalities are supported when paired with an iPhone). Moto 360 communicates with the phone via Bluetooth Low Energy (Bluetooth LE) or a WiFi network (when both phone and watch are connected to the same WiFi network) and also has the ability to independently communicate with the Internet. It can receive notifications from the phone such as reception of a phone call, SMS, email, and social media updates. Also, Moto360 has the *Google Fit* app that allows users to track their fitness activities with automatic activity recognition for activities such as walking, running, and cycling.

Moto 360 is equipped with a Qualcomm Snapdragon 400 with 1.2 GHz quad-core CPU and an Adreno 305, 450 MHz GPU [84]. It has 4 GB of internal storage and 512 MB of RAM. The capacity of the Li-Poly battery is either 300 mAh or 400 mAh based on the exact model, that is claimed to provide 1.5 - 2 days of use without charging. Display of the Moto360 is of backlit LCD type compared to the currently more trendy choice of AMOLED displays. A wide range of sensors are included in the device such as accelerometer, gyroscope, vibration/haptics engine, optical heart rate sensor, and microphone.

Major negative reviews about Moto 360 is related to its design. While the Moto 360 series is the first smartwatches to come in a circular shape resembling a traditional watch, there are some caveats. First, there is a portion of the screen that is not used at the bottom (commonly referred as *flat tyre*) [85]. This portion of the screen is kept unusable because it has a proximity sensor and some ambient light sensors. However, this appears to be a design choice rather than a technical limitation as there are subsequent smart watches such as Huawei Watch [86] and several LG watch models [87] that support the

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full circular display without the *flat tyre*. Second, the Android Wear operating system is not designed to fully support circular watches, and as a result some notifications and screens appear misaligned. Other complaints include the limited battery and inaccuracies in activity recognition [88].

Other examples of smart watch products are *Apple iWatch* [30], *Samsung Gear S2* [31] and *Pebble time* [33] and most of them have comparable feature and hardware specifications. Table XV in the Appendix provides detailed comparison of 10 popular smart watches.

While most of the smart watches have software and hardware features comparable with previously explained Moto 360, there are some unique features that stand out among the devices we surveyed. Majority of the smart watches do not support in-built GPS modules, rather rely on the GPS capability of accompanying smartphones. This can be inconvenient for the users, who want to track their fitness activities such as running as they need to carry the smartphone with them to track the path they ran and the distance. For example, smart watches: Apple Watch, Samsung Gear S2, and Fitbit Surge have in-built GPS modules, whereas other watches like Huawei Watch, Motorola Moto 360, and Pebble Time rely on the GPS sensor of paired smartphones.

Another distinguishing feature is the availability of a heart rate monitoring sensor. Again this is a feature useful for users who want to use the smart watch as a fitness tracker. Seven out of ten smartwatches we surveyed have heart rate sensors and there are two types of heart rate sensors. Smart watches such as Fitbit Surge and Apple Watch monitor heart rate continuously by default, whereas the continuous heart rate tracking function needs to be triggered using apps on watches like Moto 360 and LG Urbane 2 during a workout.

We also noticed some differences in communication capabilities of the smart watches. For all the smartwatches we studied, the normal mode of communication is to pair it with a smartphone over a short range communication technology (predominantly this is Bluetooth/Bluetooth LE). However, some smart watches required smart phones of specific types for the full operation. For example, Apple watch can be only paired with an iPhone. Also, as mentioned previously full features of Moto360 is only available when it is paired with an Android phone. While the exact reason for this is unclear, we believe this limitation occurs due to both commercial (maintaining the eco system) and technical reasons (OS compatibility). Some smart watches also had the ability to connect to Internet via 3G/4G (e.g., LG Urbane 2 [89] and Samsung Gear S2 [31]) and 7 out of 10 of the smart watches listed in Table XV support WiFi connectivity without the need of a smartphone.

Finally, beside the two large areas of smart watch applications; ***communication and notification*** and ***monitoring human physiological signals and biomechanics***, some smart watches in the likes of Samsung Gear S2 and Moto360 are also equipped with ***ambient sensing*** capabilities such as light sensors which enables them to change the screen brightness accordingly. Such features have been used for novel applications that are not currently available in consumer wearables [90] as we discuss in Section II-B.

To summarise, smart watches can be considered as a market success. As mentioned earlier over 50 million device sales were predicted for 2016, more than 50 smartwatch models are estimated to be in the market [91]. Also, more recently in addition to the manufactures coming from smartphone industry such as Motorola, Apple, and Samsung, and manufactures started directly with smart watches such as Pebble, number of regular wrist watch manufactures such as Casio, Tag Heuer, Fossil, and Rolex, are also coming up with smart watches which also indicates the popularity of smart watches. Nonetheless, the product analysis showed that there are still some usability issues mainly related to the operating system readiness such as limited support in different types of smart phones and support for different types of watch faces which more resemble conventional wrist watches. Finally, number of smart watches offer distinctive features such as GPS sensor, heart rate sensor, and 3G/4G or WiFi connectivity to the Internet.

Wrist Bands: Wrist bands are the other popular category of wrist-worn wearable devices. While there are similarities with smart watches, wrist bands are designed specifically to track a focused set of health and fitness activities. As a result, most wrist bands do not have a display screen for notifications or have a limited form factor compared to smart watches that aim at replacing the conventional watches.

UP4 by Jawbone [36] is an example wrist band that utilises bio-impedance sensors to capture signals such as heart rate, movements, body temperature, and galvanic skin response (GSR). It can passively track and record user activities such as walking and running and also has the ability of sleep monitoring. There are five bio-impedance sensors located on the inner side of the band that measure the resistance of the skin tissues to calculate heart rate. Other sensors in UP4 include a tri-axis accelerometer and temperature sensors (separate sensors for skin temperature and ambient temperature). Being a wrist band UP4 does not have a screen. The data is visible via the smartphone app that is available for both iOS and Android. Through NFC, UP4 allows to make payments like a credit card.

Similar to most smart watches, UP4 wrist band uses a battery of type Li-Poly with a capacity of 38 mAh that is claimed to sustain for 7-8 days without charging. However, we highlight that compared to smart watches, wrist bands have batteries with less capacity as they don't have large display screens. Being a proprietary platform that does not allow third parties to develop applications, details are less known about UP4's operating system and the CPUs used.

Major limitation of UP4 involves its inability to measure continuous heart rate as there is no optical heart rate sensor. Instead, it measures two types of heart rates; *resting heart rate* and *passive heart rate*. Resting heart rate is the measurement taken just before the wearer gets out of bed in the morning and passive heart rate is measured at random times during the day when the wearer is sitting still. While these two measurements give a good indication of one's general health condition, sometimes it is important to take heart rate measurements during workout sessions which is not supported by UP4 and is a limitation of bio-impedance sensors.

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Examples for other wrist bands comparable to UP4 include *Fitbit Flex* [37], Huawei Talkband B3 [92], and *MOOV NOW* [38]. We provide specifications of 10 popular wrist bands in Table XVI in the Appendix. We highlight that unlike smart watches which usually come with an operating system, wrist bands are more designed as embedded systems and some hardware details about them are not publicly available.

While majority of the wrist bands focus on ***monitoring human physiological signals and biomechanics***, we found several exceptions. For example, the *Empatica E4 Wristband* [93] is designed for research purposes and let developers to come up with innovative ideas by making the sensor data accessible. *Empatica E4 Wristband* contains most of the standard sensors such as accelerometer and heart rate sensor. However, what makes it unique is its Electrodermal Activity (EDA) sensor that measures signals related to stress, engagement, and excitement. E4 wirstband and its previous version (E3) have been used in many research studies. For instance, Müller and Fritz [94] demonstrated the possibility of classifying positive/negative emotions and low/high progress (in software development tasks) using EDA data captured by an E3 wirstband, as well as Electroencephalogram (EEG) and blood volume pulse (BVP) measured using other sensors. *Nymi Band* [39] has a different application scenario where it provides the user a simpler way of authentication. Nymi Band uses the heart rate measurements collected through its ECG sensor to authenticate users for tap-and-go payments.

Currently wrist bands are expanding their market rapidly and ever increasing interest in health and well-being will potentially ensure rapid sales growth in upcoming years. Approximately 40 million device sales were predicted in 2016 which was only second to smart watches and Bluetooth headsets. The embedded device design of wrist-bands makes sure their long battery life meets the requirement for extensive outdoor use in activities such as cycling, running and hiking. On the other hand such designs limit the interactivity with the users and the primary function of the device is to passively monitor user activities.

Overall, wrist-worn wearables are one of the main contributors of making consumer wearable products mainstream. The two main sub-categories of wrist-worns *smart watches* and *wrist bands* at the moment address two different user needs; *replacing the traditional wrist-watch and acting as an extension device for the smartphone* vs. *accurate and specialised tracking of a range of fitness activities* with some overlap in basic fitness tracking functions. These two types of products are likely to merge in future for day-to-day fitness tracking. Nonetheless, it is likely that more sophisticated fitness tracking wrist bands will continue to exist for users who need advanced analytics.

b) Head-mounted devices: Head-mounted devices are wearables worn over the head/neck area. We identify two major sub-categories within the head-mounted devices category, namely *a) Smart Eyewear* and *b) Headsets and Ear-buds*.

Smart Eyewear: Smart eyewear includes spectacles or contact lenses with sensing, wireless communication, or other capabilities such as virtual or augmented reality.

Although not expected to be worn for day-to-day use, *Microsoft Hololens* [43] is a pair of mixed-reality glasses that blends the physical world with the 3D holographic content (augmented reality). It runs the Windows Holographic platform and runs 32-bit Windows 10 operating system. Hololens has an inertial measurement unit (IMU), four "environment understanding" cameras, two ordinary cameras, one microphone array and one ambient light sensor. *Microsoft Hololens* enables gaze control of the cursor, gesture control to open apps, and drag and drop holograms, and voice commands for navigation. The example applications of Hololens includes ***gaming***, ***3D-design (e.g., architecture and home design)***, and ***remote assistance***.

HoloLens is powered by a 1 GHz, CPU, and a custom GPU for holographic processing. It has 2 GB of RAM and 64 GB of flash memory. In terms of communication, it supports WiFi 802.11ac and Bluetooth LE and can communicate with any device that support them. However, the development of the HoloLens is limited via a Windows 10 machine. HoloLens is expected to have 2-5 hours lifetime of active use depending on the load and 2 weeks of standby time [95]. The capacity of the battery bank is 16.5 Wh.

One unique feature of the Hololens is its ability to operate standalone without any companion computer or a smartphone. Major negative review of the HoloLens is, despite the adjustable headband, the device is reported to be hard to wear in a comfortable way as it is not meant to sit on the nose like traditional eyewear and most of the weight is carried by its front part. The total weight of the device 579 g which is slightly to the higher side for a wearable. Another limitation that has been highlighted is the limited field-of-view where the user's view is clipped at level that disturbs the experience. Also the active battery life is considered to be insufficient for some application scenarios.

Other commercial smart eyewear are similar to the prototype product released by Google, Google Glass (discussed in Section II-B: Research Prototypes) where the device majorly acts as a ***communication and notification*** tool. For example, FUN'IKI Ambient Glasses [44] is a pair of digital spectacles that can display phone notifications in different colours or patterns and generate sound signals from the mini speaker depending on the notification type. Being a pair of glasses supporting limited notifications functions with the assistance of a smartphone eliminates the need of having complex hardware. As a result, the device weighs only 32g. Being smart eyewear with a limited set of functionalities we could not find any major negative review about the device.

Recon Jet [45] is a more advanced pair of glasses that mainly targets long distant runners and cyclists. It allows users to get a real time summary of their activity by displaying the information such as calories burnt, pace, and activity duration without disturbing their ongoing activity. Another unique feature of Recon Jet is the ability to store maps for offline use so that the map and navigation path is visible to the athlete. It also has the ability of displaying notifications of a paired phone and is able to track the wearer's gaze and switch on and off the display automatically to save power. In addition, its built-in camera captures photos and videos instantly and helps

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to frame the shot. Another feature of Recon Jet is its ability to connect various other sensor devices for data collection.

Recon Jet is equipped with a number of sensors such as the accelerometer, gyroscope, altimeter, barometer, magnetometer and GPS. It also has a wide range of connectivity modes including ANT+, Bluetooth 4.0, and WiFi. It is powered by a Li-ion battery which claims to be sufficient for up to 4 hours of active use. It has 1GB of RAM and 8GB of flash memory and the CPU is 1 GHz dual-core ARM Cortex-A9. It runs a custom built Android version called ReconOS. The total unit weighs approximately 85 g.

One commonly highlighted problem in Recon Jet is the battery life [96], [97] where four hours might not be enough for longer activities. Also, the users have mentioned that they feel the weight of the device especially when they tilt their head during an activity such as cycling. Another potential limitation that has been highlighted is the display unit impairing the vision.

With the recent advancements of virtual reality and augmented reality, it is likely that smart glasses will become a popular wearable type in future. For example, devices such as Microsoft Hololens is expected to find further innovative application scenarios. In addition, smart glasses might become mainstream as productivity tools for enterprise environments. While as now there are only a limited set of commercial devices in this category, we expect smart eyewear to become a major category in the wearables market in next few years. Major design limitations in the current products involves limited battery life and the weight of the devices.

Headsets and Ear-buds: Bluetooth headsets are the first consumer wearable devices and still by far the most sold wearable type in the market. Traditional Bluetooth headsets are designed to make hands-free calls while more recent products are able to perform more advanced tasks (these products are also known as *hearables*) and we below discuss some of the commercial products that provide additional functionalities.

The *Sony Xperia Ear* [52] is a more intelligent version of a Bluetooth enabled headset. Apart from the typical functions such as taking calls and playing music, the ear bud is capable of taking voice commands such as navigation inquires, dictating messages, and searching the Internet. With the built-in proximity sensor, accelerometer and gyroscope, *Sony Xperia Ear* is able to tell if the device is worn by a user, not being used, or being charged. It can also detect simple head motions like a nod of the head. The ear bud works by communicating with a paired handheld device through NFC or Bluetooth and the host mobile phone app (supports only Android) allows users to make customised settings.

The device weighs only 6.6 g and can be charged using a portable charging pod which is embedded in the carrying case. The battery that powers *Sony Xperia Ear* has a capacity of 65 mAh and the charging pod has a capacity of 300 mAh. The ear bud is expected to provide 4 hours of talk time when fully charged. With the support from the charging pod, the device is expected to provide 16 hours of active time and the standby operation time is claimed to be 80 hours. Further details on the embedded hardware is not released by the manufacturer.

Main drawback of *Sony Xperia Ear* is its short battery life during active use as users claim that 3-4 hours of operation per charge is not enough especially when they do activities such as listening to music. Another limitation highlighted is the limited capability of the *artificial intelligent assistant*. For example, to trigger voice commands the user first need to tap on the device button which is inconvenient. Also, as *Sony Xperia Ear* is a one-sided earbud, it is not a good option for listening to music and the optimisations done to preserve the voice quality by eliminating the background noise also appear to interfere with the playback quality [98].

Recently released *Apple AirPods* [53] is a comparable product to *Sony Xperia Ear*. However, there are another set of earbuds that focuses on ***monitoring human physiological signals and biomechanics***, in addition to being an intelligent assistant to the smartphone such as *Samsung Gear IconX* [99], *Jabra Elite Sport* [100], and *Bragi Dash Pro* [54]. We next delve into the features of the *Bragi Dash Pro* as an example product.

Bragi Dash Pro comprises of 23 sensors including an accelerometer, a heart rate monitor, and an oxygen saturation sensor and it has the capacity of automated activity detection for activities such as running, cycling, and swimming. It consists of a 32 bit ARM processor and 4GB of internal memory, which allows users to store large amount of fitness data internally as well as storing any music they want to listen during the activities eliminating the need to carry a phone. The device is powered by a 100 mAh Li-Poly battery that can accomodate an active time of 4-6 hours and a standby time of nearly 10 days. The operating system of the device is called as *Bragi OS*. Bragi Dash Pro offers a unique feature of real-time face-to-face conversational language translation that allows two users wearing Bragi ear-buds to communicate with each other without knowing the other's language. Another feature is the possibility of using head-gestures to interact with the device which comes useful when the users hands are pre-occupied. Being released only in May 2017, so far no major negative reviews were reported. The previously known issue of the fit in the ear appears to be corrected in this version by partnering with Starkley, a company specialised in hearing care.

To summarise, we see headsets and ear-buds are redefining themselves from traditional communication and notification wearables to *hearables* with smarter functionalities and more focus on fitness tracking. The advantages of tracking fitness activities using an ear-bud is two-fold. First, the accuracy can be higher for certain types of measurements such as heart rate [101]. Second, it allows the user to receive real time non-invasive feedback as voice that can be useful in fitness activities. As a result, we expect in future the *hearables* to provide strong competition to wrist-bands as they can provide advanced artificial intelligent functions such as voice assistance and feedback, real time translation, and voice commands. However, being smaller devices fitting into the size of the ear *hearables* might not be able to support the power and processing capacities wrist bands can offer.

In conclusion on the ***Head-mounted Devices***, the current consumer market majorly consists of two types of products; smart eyewear and headsets / ear-buds. While the Bluetooth

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headsets for voice communication and music play still remain the main use case, multiple products are emerging that can track fitness activities and act as intelligent assistants. Also, the recent advancements in augmented / virtual reality also will give a strong boost to the products in this category, especially to the smart eyewear as there are number of innovative application scenarios [102], [103]. Nonetheless, design challenge of manufacturing compact and lightweight devices that can be conveniently worn while providing all the intelligent functions remains an open question. For example, these types of devices have limited battery life due to the size and weight restrictions and thus need to come up with alternative charging mechanisms.

c) Other accessories: Under this sub-category of accessories, we discuss the devices that are not mainstream enough to have a separate category as such may be worn on different parts of a human body. The functionalities of the devices are also diverse and distributed. For clarity, we present these devices in two major sub-categories, namely *a) Smart Jewellery* and *b) Straps*.

Smart Jewellery: Smart jewellery sub-category includes products such as smart rings, necklaces, and clips. As they are small in size, these devices tend to have specific, focused, and limited features compared to general purpose features provided by smart watches or wrist bands. Typically smart jewellery are designed for ***alerting users on notifications from smartphones***, ***making payments***, ***tracking of human physiological signals and biomechanics*** or ***ambient sensing***. As the products of these types are less common and quite new to the wearable market available technical information and reviews are limited.

For instance, Bellabeat Leaf [57] is a health tracking smart jewellery specially designed for women. It weighs only 18g and contains only a tri-axis accelerometer and a haptic vibration motor for notifications. Its design makes the device suitable to wear as a necklace pendent, a clip, or a bracelet. It tracks breathing, daily activities, sleep, menstrual cycles and sends results to the Bellabeat app via Bluetooth. The device does not contain any button or a display screen and no periodic charging is required. Its coin cell battery (Lithium-Manganese Dioxide) of capacity 225 mAh is expected to be sufficient for 6 months of operation. Major complaints about Bellabeat include lack of commonly used sensors such as GPS and heart rate monitor. Another limitation is that the device does not automatically sync with the app, thus users always have to do the syncing manually by tapping a button on the app.

Ōura ring [104] continuously tracks sleep and activities by sensing data from arteries in the finger. The compatible smartphone app generates reports for total sleep, rapid eye movement (REM) sleep, training frequency and amount, and evaluates overall readiness in terms of sleep balance and activity balance. The ring can tell if it is on the finger and syncs with the Ōura app seamlessly.

Another emerging smart ring is Kerv ring [56], which is a contactless payment ring funded by the crowdsourcing Web site *Kickstarter*. Currently the devices are in production and it appears that some have been released to the market, potentially to the initial Kickstarter backers. In addition to NFC based tap-and-go payments, Kerv ring is also expected to store some limited information such as emergency contacts and medical information and has the ability to use with NFC-based door locks. Being a NFC device Kerv ring does not require any battery. The NFC Ring [105] is a similar product. Being in a pre-production stage, no hardware specifications are available for either of these rings.

CliMate [106], is a clothing clip-on that tracks the ambient humidity, UV rays, and temperature. It sends data to a smartphone via Bluetooth Low Energy regularly and provides users the weather information and alert them of changes. Also, it can send alerts of when to put on sunscreen if the wearer is outdoors. CliMate operates on a coin cell battery with a typical lifetime of six months.

Smart jewellery is one area of wearables that is expected to grow in coming years and currently remains under-explored. Major advantage of incorporating sensing functions in smart jewellery is the ability to wear conveniently. Also, a synergy between the tech industry and the fashion industry can be expected and in next few years and we might see expensive smart jewellery made out of tradition material such as gold.

Straps: Chest straps, belts, arm bands (strapped onto the upper-arm in contrast to a wrist band), or knee straps equipped with sensors for health tracking or other functionalities are classified as smart straps. Smart straps find applications in ***monitoring human physiological signals and biomechanics*** and ***sensing and haptic applications***.

The *Zephyr Bioharness* [61] is a wireless physiological monitoring device consisting of an adjustable fabric chest strap and an attached Bluetooth transmitter module. The device is able to measure essential physiological signals like ECG, heart rate, breathing rate, skin temperature, and posture all in real time. It also measures respiratory rate by detecting chest expansions and contractions through an embedded capacitive sensor which is made of conductive fabric, foam, and flexible mylar. Nonetheless, this product is not designed for casual users because of its high cost as well as the complexity when accessing data. Even though raw fitness data collected at fixed sampling rates can be accessed, the whole data log of an entire day may take up to several hours to download.

The MYO armband [60] senses gestures and motions of the wearer using a range of EMG muscle sensors and motion sensors. It enables touch-free gesture based interactions with Bluetooth enabled devices for activities such as Web browsing, gaming, and presentation navigation. MYO band also provides haptic feedback in form of vibrations. The in-device processor is an ARM Cortex M4 and the strap is powered by two Li-ion batteries of 250 mAh each, that are in combination expected to provide one full day of active use per single charge. The total weight of the device is 93 g. Negative feedback on MYO armband involves the limited range of gestures it can support and the difficulties in initial calibrations.

Active Protect Wear [107] is a smart belt for the elderly, with in-built air bags that will be automatically inflated in case of a fall. The product appears to be still in the stage of development with no technical specifications available. WELT [108] (Wellness Belt) is a recent smart belt product launched in Kickstarter which is also backed by Samsung. As of now it is

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still an early stage product and is available only to the backers of the project. According to available information, the belt offers features such as tracking waist size, food intake, step count, and sitting time. The belt is claimed to support 20 days of use per charge. Being an early stage product, no further information is available about WELT.

Similar to smart jewellery the straps are still at a very early stage. None of the current consumer straps are designed for pervasive use and are still first generation products. However, there is possibility that some strap types such as smart belts emerging as mainstream products for fitness tracking or in a suite of fitness tracking products that can be worn in different parts of the body to take different measurements.

To sum up, the *Other Accessories* category contains a range of products that are in a very early stage and their impact in the consumer market is yet to be seen. According to the current state of the products, we believe smart pendents and smart belts have the potential to be a major player for fitness tracking and smart rings can find a significant market in one-off actions such as making a payment or unlocking a door. Also, similar to conventional watch manufactures moving to build some smart watch products, in future top jewellery manufactures may also enter the market.

2) E-Textiles: E-Textiles are main clothing items that also serve as wearables such as smart clothes and smart shoes. E-textiles are majorly used for: ***i) monitoring human physiological signals and biomechanics*** for various applications such as health and sports [109], [110], ***ii) ambient sensing*** for military or hazardous environments [111], and ***iii) sensory/haptic applications*** such as therapeutic massages [112]. E-Textiles are either made of conductive and comfortable-to-wear materials such as conductive fibres and flexible electronic circuits or are ordinary clothing items with conventional sensing, actuating, and data transmission modules. We classify e-textiles into two main subcategories: *Smart Garments* and *Hand/Foot-worns* and provide detailed descriptions of example state-of-the-art commercial products.

a) Smart garments: Smart garments are the main clothing items such as shirts, pants, and undergarments. There is a range of such products for ***monitoring human physiological signals and biomechanics***. For example, *Athos* [64] a Silicon Valley based start-up that produces a range of fitness gear in the likes of shirts, shorts, and capri pants [64] for workout tracking. Number of sensors such as electromyography (EMG), heart rate, and respiratory sensors are attached to these clothing items and the collected data is transmitted to a smartphone app through a hub module (named as *Core*) via Bluetooth. The app subsequently visualises the muscle effort, muscle activation, heart rate, and balance and provide feedback to users. *OmSignal* [113], *MYONTEC* [114], *Cityzen Sciences* [115] and *Physiclo* [116] also manufacture similar products.

Another type of smart garments monitor human vital signs for health, rehabilitation, and safety applications. The *WEALTHY project* [117], Georgia Tech Wearable Motherboard [118], and ProeTEX [119] are early examples of wearable health care systems that utilises knitted fabrics to monitor essential physiological and bio-mechanical signals, such as heart rate, breathing rate, and dehydration. More modern consumer oriented products include *Exmobaby* [120], *Cyrcadia Health* [121], *B Maternity* [122], and *WEMU* [123]. For example, *Exmobaby* is a cotton baby pyjama with a removable strap on the inside that contains fabric to measure baby's ECG. In addition, the pyjama also contains a thermometer and motion detection sensors externally. The data is transmitted to a bay station over FM and the bay station subsequently transfers the data either via WiFi or 3G. *Cyrcadia Health* has developed *iTBra* that uses thermal dynamic measurement to detect breast tumours. Changes in body temperature are picked up by sensors in the bra is transferred to the cloud and the analysis is forwarded to a smartphone [124]. Similarly, *B Maternity clothing* is a prototype product developed to monitor pregnant women and *WEMU* is a top designed to monitor epilepsy patients.

Some other smart garments have ***ambient sensing*** functionalities for the safety of the wearer or as a means of data collection. For instance, *Ohmatex* [125] and *Viking Life* [126] introduced a sensor embedded firefighter jacket. With the thermal sensors embedded directly to the inner and outer fabric layers and LED displays, the outfit gives the advantage to visually indicate critical heat levels and potentially critical situations. A battery in the liner is used to power the whole system. *Safe@Sea* [127] is another example suit for fishermen with integrated sensors that emits an emergency distress signal, if the person wearing the suit falls into the water.

Sensory/haptic applications are another major use case of smart garments. The *Hug Shirt* [65] enables people to send hugs over distances. The embedded sensors detect the touch, warmth, and heart rate of the sender. On the receiver side, the actuators recreate the sensation of touch, warmth, and emotion of the sender. Similar to the idea introduced by Hug Shirt, a more recent E-textile product *Tjacket* [128] is a wearable tech vest that is designed for children with ADHD or Autism who are sensory seeking and/or sensory over-responsive. Using air pressure to simulate hugs it provides deep touch pressure for which the levels, durations, and times can be controlled via Bluetooth using a smartphone. Parents and caregivers can then monitor the effects of the hugs via an online service. Currently the price of such a vest ranges from $399-$599, which is much higher than regular children clothes.

The *WarmX* [129] is a heatable underwear for sufferers of muscular tension and back pain. The silver-plated polyamide threads of low resistance are woven into the underwear to conduct current and to warm up directly on the skin. A small battery control unit is connected to the conductive fabric to power the heating zones. *Spinovo* [67] is a similar prototype product of an upper body garment for back pain relief.

Apart from the above main use cases, there are smart garments for a range of other activities. The *Solar Shirt* [66] takes use of flexible solar panels to harvest solar energy and stores the energy in a battery. The circuit embedded on the shirt is able to provide a suitable charging current for other handheld electronic devices. *Samsung's NFC smart suite* [130] and *Lyle and Scott contactless jacket* [131] allow to make card-less payments and exchange business cards.

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b) Foot/hand-worns: Other subclass of E-Textiles is smart foot/hand-worns such as smart shoes, smart socks, and smart gloves. Similar to smart garments, smart foot/hand-worns also find application scenarios in ***monitoring human physiological signals and biomechanics***.

Sensoria [71] is a pair of sensor integrated socks designed for runners to avoid injuries. Three textile pressure sensors under the plantar area monitor the wearer's foot-landing techniques. The bluetooth enabled anklet contains a 3-axis accelerometer to enhance data monitoring and utilises Bluetooth for data communication with the corresponding smartphone app. The anklet is mounted to the sock and the specially-designed charger by magnetic connection points. Even though developers have tried to make these socks as conventional as possible, there still exist some negative reviews about the usability. Both the textile sensors and the anklet that weights 27.2 g are said to be noticeable and slightly uncomfortable when first putting on and while standing [132]. Footwear manufacturer *Nike* was recently granted a patent for a sneaker that has an embedded fitness tracker in the sole [133].

In addition to fitness applications foot/hand-worns also target health and personal safety applications. For example, *Owlet Smart Sock* is a sock for infants which is equipped with a four-sensors pulse oximeter and monitors babies in sleep and alerts the parents if the child stops breathing, experiences a drop in blood-oxygen levels, or has irregular vital signs. The sock is powered by a rechargeable coin-cell battery and communicates with the smartphone app over Bluetooth or WiFi. *Edema* [134] is a wearable clothing item for the calf muscle, that contains a strain gauge that measures the changes in leg volume. It provides detailed information about fluid accumulation and excretion rates for patients suffering from congestive heart failure and pre-eclampsia.

Shoes and other foot-worn devices are also suitable for purposes like location or distance tracking by nature. *Lechal* shoes [70] measure steps, distance and are able to alert the visually impaired users on nearby obstacles with an adjustable vibration. *GPS Smartsole* [135] is a pair of shoes with a GPS tracking module built into the sole allowing to track the wearer's location and the product is mainly intended for young children or sufferers of Alzheimers and Dementia.

While we did not come across with any *Foot/Hand-worn* for ambient sensing, there are number products for ***sensory/haptic applications***. For instance, Fujitsu released a prototype product of a *gesture-control glove* that provides touch and gesture-based functionality for industrial site operations [72]. The glove is targeting workplace environments where the wearer needs to keep both hands free or need to avoid touch based devices getting damaged or dirty. The device has a contact sensor on the finger, and near field communication (NFC) tag reader allows users to touch the NFC tag of an object with which they are working in order to digitally identify it. *Goldfinger* is a similar prototype [136].

Compare to *Accessories* category, the availability of products in *E-textile* category is limited and still not adapted significantly by the consumers. One of the major areas where E-Textiles are making significant inroads are in baby products, where their clothing items are used to give vital information such as breathing rate and blood-oxygen level to the parents wirelessly. In addition, there are also some products in fitness tracking, especially the smart shoes and soles that are capable of tracking some metrics such as step count more accurately. Overall, there is still room for further exploration in E-Textile as textiles covers a significant part of the body all time and have the potential of collecting number of health related data. However, usability becomes a major significant factor as textiles need to be comfortable to the wearer even more so compared to accessories.

3) E-Patches: We define *E-Patches* as patches that can be adhered or tattooed to the skin. *E-patches* can be classified into two major categories; *Sensor Patches* which are small devices embedded with sensors and *E-Tattoo/E-Skin* which are skin-like e-patches integrated with miniaturised sensors and circuits. A large number of *Sensor Patches* are made for ***monitoring human physiological signals and biomechanics***, while others are designed for ***Sensory/haptic applications***. *E-Tattoo/E-Skin* have a wider range of applications because of their unique characteristics. A closely related product type to *E-patches* are the *Implantable E-patches* which are placed under the skin. Nonetheless, such products are not considered in this survey as the focus of this survey paper is on wearable devices for daily use.

a) Sensor patches: Sensor Patches designed for ***monitoring human physiological signals and biomechanics*** usually consist of reusable sensor modules with disposable and adhesive patches. For example, the *HealthPatch MD* module by VitalConnect [137] facilitates continuous monitoring of essential vital signs of a human body by detecting electrocardiography (ECG) and acceleration signals. Respiration rate is extracted from the ECG signal and tri-axial accelerometry. The sensor patch can also monitor body temperature, postures, and detect basic activities of the wearer. The patch is easy to use and the users have to peel off the adhesive protector at the bottom. A similar device, *e+ Emergency Patch,epatch* is specifically designed for situations where medical attention is not accessible, i.e., during a flight. This device includes an ECG sensor, a respiratory band, and a pulse oximetry sensor. The patch is coupled to the chest and the pulse oximetry sensor is placed on the finger tip. *Proteus Health Patch,proteus* is a similar product. Besides the regular health tracking capabilities, the patch is also able to track medication-taking of their special sensor embedded pills.

Some other *Sensor Patches* are designed for ***sensory/haptic applications***. For example, *Thync* [77] is a strip providing relief to the wearer to lower the stress levels. It generates low-level electric pulses to stimulate nerves on the neck and head to alter the brain's Adrenaline release. Another example, *UPRIGHT* [78], an e-patch designed for posture training, notifies a wearer in real-time using vibrations when he slouches. The device communicates with a customised smartphone application through Bluetooth to log training progresses.

Besides the two application scenarios described above, which are also seen in other product categories, one special use-case of smart patches is drug-delivery. *The OmniPod Insulin Management System* [140] is an insulin pump therapy

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developed for Type 1 diabetes. Being an award winner, the system has remained competitive in the market for many years [141]. The system consists of two parts: the OmniPod Disposable Infusion Pump (Pod) and the OmniPod Personal Diabetes Manager (PDM). After the filling process, the pod injects insulin to the subcutaneous tissue through a soft cannula integrated underneath. The pod also incorporates a blood glucose meter. The injection of insulin and the collected glucose data are managed on the handheld PDM. The devices communicate via radio frequency signals and need to be within 60 cm of distance. *Chrono Therapeutics* [142] designed a patch for smoking cessation. The patch tracks the personal use of medication and deliver a proper amount of Nicotine Replacement Therapy (NRT) to the user.

b) E-tattoo/E-skin: E-patches can also come in the form of temporary tattoos or electronic skin. These tattoos take use of flexible and stretchable electronic circuits to realise sensing and wireless data transmission. Functions such as ***monitoring human physiological signals and biomechanics***, ***communication and user notification,*** and ***making contactless payments*** can all be found in different existing products.

One example is the *Motorola patented e-tattoo* [82], that is integrated with tiny ECG, EMG, strain gauge, temperature sensors, and a microphone. According to the available information, audio signals can be picked up from the tattoo, and converted into digital form. The voice signal can be used to unlock the wearer's mobile device. L'Oreal's *My UV Patch* [143], is a tattoo made of stretchable electronics that can measure the amount of UV exposure of the wearer.

The *Wearable Interactive Stamp Platform* [83] is an ultrathin, tattoo-like stamp designed by *MC10* and *PCH*. Paired with an NFC enabled device, the smart stamp enables wearers to make card-less NFC payments, access a hotel room or transmit health information.

Compared to the traditional sensor patches and sensor embedded textiles, skin-like E-Patches are more versatile, less noticeable, and more comfortable. Most of the e-skin products are still at a very early development stage, and with the advances in flexible electronics, we can expect to see more interesting application scenarios and products in this category. We finally highlight that as the products are in early stages, detailed hardware specifications are not yet publicly available.

B. Research Prototypes

Under research prototypes, we survey the recent academic research on novel wearable devices and also recent industrial attempts, e.g., start-up products that are not yet commercially available, utilising the existing state-of-the-art hardware and software capabilities as building blocks for new devices and services. We adhere to the same classification as the existing products section, i.e., **i) Accessories**, **ii) E-Textiles**, **iii) E-Patches**, and all products are categorically compared with existing products in Table I.

1) Accessories: Majority of research prototypes is focused on one specific application or a small set of functionalities where as existing accessories presented in the previous section are more often general purpose devices that provides wide variety of services. In a way, the research community leverage the existing general purpose accessories to demonstrate the true potential of wearables with novel services. In this section, we review the recent work related to *Accessories* under three sub-categories **wrist-worn**, **head-mounted** and **other accessories**.

a) Wrist-worns: Existing products in wrist-worns such as smart watches and fitness bands are the most popular and well refined in terms of hardware capabilities. Therefore, recent research works in this category are primarily on augmenting the services offered by the exisiting devices.

Smart Watches: Gait-Watch [144] is a context-aware user authentication system based on the gait pattern recognition from a smart watch. The authors propose a novel sparse fusion method to improve the accuracy and the complete system is implemented on a Samsung Gear Live smart watch for the validation. The results show that Gait-Watch authenticate the person with 95.2% true positive rate. Similarly, Yang *et al.* [145] proposed a motion based authentication for wrist worn devices with IMUs. Simple histograms of accelerometer signals and dynamic time warping are used to for feature extraction and then the similarity score of features are used for authentication. The system is implemented on a Android smart watch. Although these systems show promise, they are not yet ready for commercial secure authentication due to probabilistic nature of these mechanisms. However, due to small screen and limited user interaction, continuous and implicit authentication mechanisms seems to be more practical approach for smart watches. Along with the advanced sensing capabilities, smart watch has the potential to be used in number of continuous and implicit authentication mechanisms in future.

Gruenerbl *et al.* [34] designed a smart watch application that assists untrained people with performing Cardiopulmonary Resuscitation (CPR). Using the acceleration data on the watch, the application gives feedback on frequency and compression depth to guide the aider. It was found that the ratio of participants who managed to stay in the ideal depth and frequency ranges was much higher for those with the assistance of the application. Xu *et al.* [35] showed that the finger, hand, and arm gestures can be accurately identified by monitoring movements of the tendons passing through the wrist using an accelerometer and a gyroscope attached to a smart watch. Specifically, the authors built a classifier that can recognise the characters written by the index finger on a surface with 95% of accuracy. Extending this work to detect continuous writing with multiple words and writing in the air is left as a potential future work. Even though the wearables such as smart-rings can be used to detect finger writing more accurately, wide availability of smart watches and fitness bands makes this work more deployable than others.

Wrist Bands: In the recent past, researchers have been trying to take wrist bands beyond the health and fitness monitoring devices. These activities can be categorised in to novel authentication mechanisms and complex activity recognition. Wearables are now driving the research on continuous and implicit authentication mechanisms because wearables are designed to be worn through out the day. To this end,

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wrist bands are the most preferred device category as they can be worn during sleeping and bathing where most other devices are not available. Moreover, wrist bands contain as much resources as smart watches, but an interactive display. Cornelius *et al.* [40] introduce new hardware element, bio-impedance sensor, for wrist bands for accurate implicit user authentication. The bio-impedance sensing system includes an impedance analyser (AD5933) controlled by Shimmer via the I2C bus and 8-channel multiplexers (ADG1608) for selection of multiple electrode locations. Four features were extracted from bio-impedance and then a Support Vector Machine based machine learning classifier is trained to uniquely identify individuals. Although it was reported that 98% of accuracy in user identification, the robustness of this method is still questionable as there are many factors such as wrist orientation, skin temperature, and diet that affect the bio-impedance. Ohtsuki and Kamoi [146] proposed another implicit authentication mechanism from the photoplethysmographic (PPG) sensor data collected from a wrist-worn device. The authors used PPG signal peaks to detect hand movements in addition to heart rate information that is naturally embedded in PPG sensor data. Based on experimental data from 15 users, the authors demonstrated the feasibility of the mechanism with 8.8% minimum error rate. However, these authentication works [40], [146] are at very early stage similar to smart watch based authentication frameworks.

Wrist bands are ideal devices to detect complex hand gestures as inertial measurement units (IMU) are a commodity in any wrist band. To this end, Parate *et al.* [41] propose a machine learning algorithm to detect smoking gestures using a wrist-worn 9-axis IMU consisting of an accelerometer, a gyroscope and a magnetometer. The set of proposed trajectory-based features are able to reliably distinguish smoking gestures from other hand-to-mouth gestures such as eating and drinking. The results show 91% precision and 81% recall in detecting smoking gestures. Watanabe *et al.* [42] present another activity recognition method that requires the user to wear ultrasonic speakers on the wrist and a receiver on the chest/neck. The activities are recognised by the volume of the received sound and the Doppler shift at the neck-worn receiver. The authors claim that the ultrasonic based recognition is able to detect activities of surrounding people with out the need of communication with other devices or data fusion. However, the performance of this device may be affected by ambient environmental conditions. The authors claim that if additional speakers can be fixed in the surrounding, the system can be extended to location recognition. Another recent work attempts to monitor hand gestures using a fabric wrist band that contains polymeric strain gauge sensors [147]. In the prototype version, the sensor data is collected via a USB connection and then analysed on a computer using common machine learning techniques such as LDA and SVM. The results were validated with more accurate gesture identification with force sensitive resistors attached to finger tip and the palm.

It appears that researchers are experimenting with new types of wrist-worn devices. However, it is more interesting to compare the results of these new devices with commercially available wrist bands, which has not been done in majority of the cases. In general, research prototypes of wrist-worns are designed with a specific purpose in mind, when the proposed methods become mature enough, we can expect to see new functions or new devices as commercial products.

b) Head-mounted devices: This is the category of wearables with the highest number of research prototypes. It can be expected because commercial head-mounted wearables are not yet mature enough compared to wrist-worns. In this section, we review the head-mounted devices that are developed using existing products like glasses while others are prototypes built with off-the-shelf sensors.

Smart Eyewear: Despite the enormous publicity received for the launch of *Google Glass* [46], still it remains as a research prototype that is not yet available in the consumer market. However, the initial release of prototypes immensely contributed to the research and development activities in the category of head-mounted devices. In a nutshell, Google Glass is a pair of glasses that allows user interaction through a touchscreen and voice control and also provides the ability of capturing photos and videos. It contains sensors similar to smart watches such as accelerometer and gyroscope and runs a version of Android operating system. Google Glass is powered by a battery of Li-Poly type with a capacity of 570 mAh. A number of design issues were raised by the product reviewers as well as academics who experimented with the limited devices released. For example, many users complain about the battery life. The glass was able to record only 30 mins of video without charging and most users considered this insufficient. Also, when in operation the glass produces heat that was uncomfortable for the users [148]. User friendliness in GUI interactions was another problem. Apart from these technical drawbacks, Google Glass lacked a *"killer application"* and sparked number of legal discussions of wearing it performing certain activities such as driving [149] and walking in public places [150]. As a result, Google Glass never became a mainstream product. Nonetheless, it attracted a significant interest from the research community to come up with innovative applications.

Google Glass has been extensively used to improve surgical procedures. Zhang *et al.* [151] exploit the capability of connecting external sensor data to the Google Glass for real time data processing and visualisation. They developed a system of hardware and software not only to visualise, but also to control the sensors and actuators by the control commands issued via the Glass. On the other hand, there are user experience and suitability analysis of Google Glass capabilities in surgical procedure and in general medical workflow [152]–[154]. Medical personnel provided positive feedback to these types of devices which suggests that it could be a reality in the near future along with the technological advancements. *Chroma* is an augmented-reality application developed using Google Glass to support colour-blind people [51]. Chroma is designed as a solution for all types of colour blindness, and it has four different modes, among which the highlighting and overlaying modes can successfully help wearer to distinguish colours. Under the current hardware configuration of the Google Glass, the app consumes considerable amount of power and can

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only last for an hour without charging. Since the current Google Glass does not support calibration of the camera, the performance of *Chroma* is also limited. However, the proposed methodology and algorithms demonstrate the potential applications of future smart eyewear. The study presented in [50] is another wearable system to support blind people. It was found that *masking* is a harmful interference in existing Electronic Orientation Aids. In this study, researchers carried out empirical studies in two areas, i.e., the understanding of masking and the usefulness of wearable devices to a blind user. Results showed that wearables combined with vocal feedback mechanism which notifies users can successfully avoid masking. However, the current context is limited by the large number of audio notifications that may be distracting. Two prototypes proposed in this paper are smart eyewear based. The first is an eyewear with cameras mounted above the lenses. It is used together with a customised glove and a belt embedded with speakers and vibration motors to send feedback. The second is a pair of smart glasses with a camera placed between the lenses.

Gaze tracking is another explored area using smart eyewear. Leelasawassuk and Mayol-Cuevas [48] presented a hand-free and feedback-free way of creating 3D models of objects by analysing the gaze data of a user collected from an eyewear. The gaze data are classified into fixations (stationary) and saccades (fast eye motion), and the outliers are eliminated when recovering the 3D model of the object. Compared to the conventional way of 3D modelling, which combines the selection of specific points and visual feedback on a computer desktop environment, this approach need much less explicit user input and it can be implemented on a wearable platform. To address the problem of high power-consumption in real-time eye-tracking, Mayberry *et al.* [49] proposed iShallow. iShallow saves energy by sub-sampling the pixels. It utilises a multi-layer neural network learning algorithm that is robust against distortions in training dataset, and includes a real-time gaze estimation algorithm for eye tracking. Compared to other commercially available products, the system consumes less power and supports mobility.

Moving forward from smart glasses, Google also recently published their work on a *Google contact lens* [47] that contains a wireless chip and a miniaturised glucose sensor embedded between two soft layers of lens material. A tiny pinhole in the lens allows tear fluid to seep into the sensor for blood sugar level analysis. A wireless antenna in the size of human hair acts as a controller to communicate information with other wireless devices in the vicinity. Smart contact lens eliminates many criticisms of Google Glass and positions it as a day-to-day wearable device. However, the true potential and user experience issues of smart contact lens are yet to discover.

Headsets and Ear-buds: In [155], a hat embedded with ultrasonic sensors and pressure actuators was designed for navigation by providing wearers with spatial information of the surrounding area. This study demonstrated that low-frequency pressure signals can be used as a replacement for vibrations that are annoying to some users. The final prototype was tested in five tasks that focused on orientation and navigation. Major issues with the prototype were related to sensor placement and latency.

In summary, smart eyewear dominates the research prototypes in the head-mounted category, primarily due to the availability of somewhat mature hardware, e.g., Google Glass. Despite number of technical and experience issues of Google Glass, it has paved the way for number of interesting research directions. Along with the increasing popularity of augmented and virtual reality applications, head-mounted devices could be the next wave of wearable devices to gain commercial success.

c) Other accessories: In this section, we present wearable accessories that are designed to wear in other areas of the human body. Similar to *Existing Products* section, we categorise them as *a) Smart Jewellery* and *b) Straps*.

Smart Jewellery: Nirjon *et al.* [58] developed a wearable ring (*TypingRing*) that enables text inputs without a keyboard. A user wears the ring in the middle finger and types using three fingers (index, middle, and ring) on an imaginary keyboard on a surface. In addition to the key recognition, *TypingRing* communicates with the computing device in real-time to provide visual feedback to the user. The experimental study showed that *TypingRing* provides an accuracy of 98.67% and 33-55 keys per minute typing speed. Gummeson *et al.* [59] propose a remote gesture input solution using a wearable ring, which utilises a finger-tendon pressure-based solution to detect touch and an audio-based solution to detect motion on a surface. Additionally, the proposed wearable ring opportunistically harvest energy from an NFC-enabled phone and supports 10 hours of active user interactions without explicit charging.

Straps: The *PneuHapic* is an early stage prototype armband developed by He *et al.* [62] that generates sensations such as tapping, holding, and tracing through air inflation and deflation. The system consists of an array of five air chambers controlled by a 5V Arduino Pro Mini via two air compressors. The stimulus used in this study was highly identifiable. However, the resolution is limited by the current topology of airflow. Since similar information can be obtained by existing sensors on current devices such as IMUs, the adaptability of this solution highly depends on the user experience, which is yet to be discovered. It is interesting to see a comparison of performance with MYO armband [60] that provides similar information with different sensors which is currently available in the market.

BodyBeat [63], is a neck-wear device developed for sensing and classifying non-speech body sounds including eating, drinking, breathing, laughter, and yawn. The sounds are captured using a customised piezoelectric microphone. BodyBeat communicates with an Android smartphone using Bluetooth. A linear discriminant classifier is developed extracting number of frequency and time domain features. Overall, the system demonstrated an average classification recall of 71.2% with low computational and energy cost. With improvements in the wearability, the system can potentially be deployed for food journalling and illness detection.

To sum up, the research prototypes presented in this section show interesting user cases, but they are not yet strong enough to gain much wider attention to progress beyond research prototypes. There can be large variety of use case ranging

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from industrial automation to augmented or virtual reality. The whole eco-system for these types of applications are currently evolving and therefore the significance of these devices will be more prominent in the near future.

2) E-Textiles: E-Textiles are an emerging category of wearables with applications in many disciplines such as health care, fitness, precision sports and health and safety. There are number of material science research activities in developing conductive fabrics and embedding sensors on fabrics which is not the focus of this section. We aim to survey research prototypes that leverage these smart fabric and wearables that are closely related to textiles such as foot-wear. We categorise them as *i) Smart garments* and *ii) Foot / Hand-worn* devices.

a) Smart garments: Google Project Jacquard [156] is aimed at developing smart garments with conductive yarns and embedding electronics into garments. The idea is to capture touch and gesture data and wirelessly interact with the smartphone. Google has partnered with the fashion trademark Levi's [156] to develop a Commuter Trucker Jacket specifically targeting the urban bike commuters. The product is expected to be released in 2017. *SimpleSkin* Shirt [157] is a shirt made up of combining smart fabric with normal fabric that has embedded sensors for capacitive sensing, resistive sensing, and bio-impedance sensing. The prototype is still at very early stage where sensors are simply attached to a normal shirt. All three sensor packs are connected to sensor boards and they connect with each other via Bluetooth. An Android app is developed with a set of APIs for developers to interact with the sensors and collect data. It also appears that the authors do not have a specific application scenario rather they are suggesting few such as continuous heart rate monitoring, gaming and implicit stress detection.

MyoVibe [68] is a wearable mechanomyography (MMG) system that was developed to detect the activation of skeletal muscles. The system consists of a set of IMUs that can be connected to any garment and a data aggregator that either saves data on an SD card or sends data to a back-end computing device through USB or WiFi. The obtained data were processed in segments of 500ms and five frequency domain features were studied over three types of exercises. A decision tree was built through supervised machine learning processes, and the results of activation prediction was checked against the manually processed sEMG (surface electromyography). The system demonstrated an accuracy of 98% and 80% when detecting isometric low motion activations and activations in high mobility/noisy environment respectively.

There are also a number of prototype level smart garments under the category of *eco-conscious fashion* that act as indicators of the ambient air quality level such as *CO_2 Dress* [158] and *Warning Signs* [159]. CO2 Dress [158] contains a set of LED lights and carbon dioxide sensor where the lights lit up according to the carbon dioxide concentration in the surroundings. Similarly, Warning Sings [159] lights up a pattern when carbon monoxide concentration is high, i.e., when someone is smoking, in the surroundings.

In general existing smart garment products are not open to app developers compared to other wearable devices. As a result, the amount of research activities that leverages technical capabilities of existing devices are also limited. Moreover, in addition to technological researchers, fashion researchers has begun to start experimenting with smart garments. Since this is an industry driven by fashion designers, this trend may take the smart garments to the consumer product level.

b) Foot/hand-worn: Goldfinger is a prototype human-machine interface glove [136]. It utilizes piezo-electric transducers attached to fingers to convert mechanical power generated by fingers into electricity to charge the glove battery, saving the need for wires and bulky power supplies. The idea is to use the glove to operate and control machines with gestures and the authors envision applications in virtual reality and medical industries.

Jain *et al.* [73] proposed a shoe-based sensing and alerting system for the safety of distracted pedestrians. The system consists of inertial sensors mounted on the shoes, two Android applications running on a smartphone for data acquisition and data processing, and a notification feature which alerts the user. Similar functioning was achieved using GPS in previous studies. Nonetheless, GPS cannot recognise a transition from the sidewalk to the street. In this study, inertial sensors (i.e., accelerometer and gyroscope) were used to detect ramps and curbs and a threshold-based algorithm was developed for event classification. The method demonstrates a detection rate higher than 90% with the false positive rate being 0.7%. Aminian *et al.* [74] investigated the correlation between several foot clearance parameters and gait speed using shoe-worn inertial sensors. The gait spatiotemporal parameters were obtained by double integration of acceleration and trajectory data, and the foot clearance parameters were calculated using the 2D kinematic model described in [75]. Through principal component analysis, weak to moderate correlation was observed in toe clearance trajectory and heel strike pitch angle.

Fashionability has always been a barrier to commercialisation of *E-Textiles*. It is impossible to commercialise any e-textile that is not fashionable. Also the textile must be durable and washable. This is clearly evident from the partnership of Google with a fashion trademark. It worths noting that majority of recent E-Textile solutions have given significant attention to make sure the product is washable and/or durable even at the research stage. However, the research prototypes in this section are far from real deployments. E-Textile research will require to collaborate more with the fashion industry to make e-textiles a commodity.

3) E-Patches: E-Patches are the most recent category of wearables that are not yet well established as consumer wearables. Similar to existing products, we classify them as either *i) Sensor Patches* and *ii) E-Tattoo / E-Skin*.

a) Sensor patches: In [160], authors developed a stick-on patch that monitors ECG and EEG signals. The device can be charged wirelessly and transmit data to a computer in real-time. This patch is well-coupled to the skin which allows patients to move and behave naturally during the monitoring period. Another device, *Electronic fingertip* [161] can sense stresses and strain at the finger tip and assess electrical properties of tissue to help doctors remove the tissue precisely by local ablation. This application can be potentially used to create electronic surgical gloves.

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b) E-tattoo/E-skin: DuoSkin [79] is a fabrication process of on-skin user interfaces. The gold leaf was chosen as the key material for its accessibility, stretchability, and comfortableness when attached to the skin. Three functionalities, namely sensing touch input, thermochromic display and communicating wirelessly with other devices via NFC can be achieved in this fabrication process.

A research group from the University of California recently developed an alcohol sensing device using flexible wireless circuits [80]. The iontophoretic-biosensing-based wearable patch measures the Blood Alcohol Concentration (BAC) from the sweat and sends data to a computing device via bluetooth. The biosensor demonstrated high selectivity and sensitivity to ethanol in this study.

Apart from the skin-contact patches, a group of researchers at National Taiwan University designed a *Smart Tooth Patch* [81] which is able to detect activities such as chewing, drinking and talking with an accelerometer and classify the activities as they move teeth differently. This will another potential solution for food journaling which is one of the critical issues in health and fitness monitoring. However, wide adoption of such are yet questionable as it is not tested with a large population.

Even though E-skins have attracted lots of interests, most of the E-skins are still in the early development stage and not yet to be commercialised. Health-care sector is currently driving the growth in E-Patches. Similar to E-Textiles, existing products and associated software packages are proprietary to the manufacturers. As a result, current devices do not drive further research activities in contrast to smart watches and smart eyewear.

C. Summary of the Product Survey

To summarise, in this section we surveyed exisiting consumer wearables and advanced research prototypes by classifying them into three categories; *Accessories, E-Textiles, and E-Patches*. We identified major applications of wearables includes ***communication and notification***, ***human physiological signals and biomechanics***, ***ambient sensing***, ***sensory/haptic applications***. In addition, our survey on advanced research prototypes showed that significant amount of research work happens on ***gesture detection*** and ***gaze tracking*** as well. Table III in the Appendix summarises all the devices and prototypes we surveyed according to their device classification and application domain. Tables IV–XIV in the Appendix provide further details on the products surveyed under various categories.

This comprehensive product survey depicted that modern wearables are sufficiently equipped with advance sensors to measure variety of human attributes as well as the surrounding environment. However, current commercial wearables and research prototypes are still struggling with ***i) Security***, ***ii) Energy*** and ***iii) Computing power***. In light of this, we organise the remainder of this survey to further investigate these three aspects of wearable devices.

In Section III, we discuss the research work that highlight ***security attacks and vulnerabilities*** of wearables and solutions proposed, in particular to communication security as it has been identified as the weak link in terms of device security. Summary statistics show that majority of the products are still dependent on a resourceful partner device and rely on Bluetooth or Bluetooth LE for communication purposes. While WiFi was available on a limited number of devices that was mainly used to communicate to the smartphone than direct Internet. Direct Internet connection via 3G/4G is only supported on a limited number of smart watches. Only LG Urbane 2 [89] and Samsung Gear S2 3G [31] smart watches and the wrist band Gizmopal 2 [162] supported a cellular connection. Common use of Bluetooth for wearables raises the question whether it is a suitable protocol for wearables as there are some known attacks. Also, the limited CPU power and battery constraints might limit the implementation of advanced security protocols.

In Section IV, we further discuss ***battery technologies and energy harvesting*** as means of achieving energy efficiency. For some of the products the major negative reviews were related to the limited lifetime of battery, making battery lifetime an important aspect of designing wearables. In terms of energy sources, majority of smart watches and wrist bands surveyed use rechargeable batteries such as Li-Poly batteries or Li-ion batteries, while some of the other products use replaceable coin-cell batteries. Some research prototypes [59], [136] have already identified the requirement of alternate energy sources such as energy harvesting from human motion or ambient environment.

In Section V, we discuss means of ***computation offloading and in-device machine learning*** techniques that are required to provide enhanced software features. We noticed that majority of the wearables are same in terms of sensor capabilities apart from the GPS and the heart rate sensor. Thus, the major differentiations in products come as software features that depend on the available CPU capacity and other methods of doing heavy computations to introduce innovative features.

Wearables often transfer collected sensor data to either nearby smartphone or directly to a cloud server for further data analysis, real-time notifications, and visualisation. Majority of wearable sensor data is highly personal and uniquely represent the user. As a result, secure transmission of wearable sensor data is an utmost important aspect in wearable systems. Due to the resource limitations of wearables, communication security is often questioned by researchers and is an area of active research. To this end, we first survey the potential security threats to current wearable communications and then review proposed solutions in the next section.

III. Communication Security

To offer a diverse range of applications, wearables rely on a wide variety of wireless communication protocols supporting different communication ranges as illustrated in Figure 3. Because wearables often need to transmit highly private information, be it health, financial payment, or any other private transactions, they must guard themselves against any security risks involving wireless data access. Unfortunately, resource constraints in the form of limited battery, CPU, memory, and

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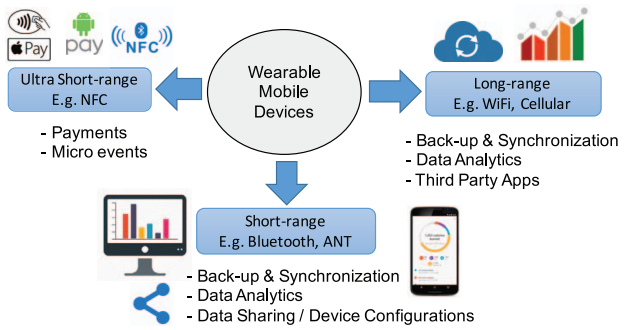


Fig. 3. Communication modes of wearable devices.

device form factor often limit the implementation of advanced secure communication measures. As such, it is imperative to identify any potential security holes in wearable wireless communications and develop practical safe guards against them. In this section, we first survey security vulnerabilities identified in the literature under the context of wearables. Second, we discuss various approaches proposed in the literature to address the identified vulnerabilities.

A. Security Threats to Wearables

We present work that highlighted security threats to wearables under three categories: ***1) Threats to Confidentiality, 2) Threats to Integrity,*** and ***3) Threats to Availability***. This classification is standard in network security [163] and has been used in multiple other studies [164]–[166]. *Threats to Confidentiality* involves the scenarios where attackers get unauthorised access to information using techniques such as eavesdropping the wireless channel. *Threats to Integrity* includes the cases where attackers alter resources or information without authorisation. *Threats to Availability* are the situations where attackers act to deny services to the entities who are authorised to use them.

1) Threats to Confidentiality: As our product survey highlighted, majority of existing wearable devices use Bluetooth Low Energy (BLE) as the major means of communication. However, BLE equipped wearable devices have been reported to be vulnerable to attacks that impact the confidentiality such as *eavesdropping* [167]–[170] and *traffic analysis* [167], [171], [172]. Another body of work [173]–[175] reported scenarios of using wearable devices to *gather information* about user's other devices such as her PIN to unlock the smartphone. We below discuss existing studies that reported and analysed these attacks threatening confidentiality in wearable communications.

a) Eavesdropping attacks: Eavesdropping is the unauthorized real-time interception of a private communication and these attacks are significant security threats to wearable systems as they can expose user's personal information to an attacker. Also, they may end up being entry points for other forms of attacks. As mentioned in Section II majority of wearable devices are utilising BLE communication protocol that can suffer from eavesdropping attacks in some scenarios. In particular, there are specialised hardware such as Ubertooth [176] and Bluefruit LE Sniffer [177] that allow adversaries to detect and follow the communication chain of a BLE wearable device.

Fawaz *et al.* [168] reported number of issues related to the implementations of the BLE advertisement process in consumer wearable devices. More specifically, authors found that some of the existing wearables do not have a proper implementation of the MAC address randomisation defined in the BLE specification as a privacy preserving provision. For instance, it was found that some devices do not use address randomisation at all whilst some implementations were insufficient as they were only flipping few bits of the public addresses. Also, there were concerns in the duration a random address is kept active. These implementations enable adversaries to eavesdrop on the wireless channel and follow the advertisement packets to track a BLE device easily.

The privacy leakage from the advertisement was also identified in other studies that involve security analysis in fitness trackers [169], [170]. For example, Goyal *et al.* [169] identified the key privacy and security vulnerabilities of the two most widely adopted wearable fitness trackers, *Jawbone UP Move* and *FitBit Charge*. To analyse the security of Bluetooth LE communication, the authors set up a Bluetooth protocol stack *bluez* and the *GattTool* tool to contact and interact with the tracker and discovered that both trackers have not implemented address randomisation. Thus, the attacker can track the user over time by eavesdropping on the device's communications and exploiting the public device address.

Similarly, recent Open Effect Report 2016 [170] provided a comparative analysis of the privacy and security features of fitness tracking devices. The authors investigated Bluetooth LE privacy provision in number of fitness tracking devices such as Fitbit Charge HR, Jawbone UP 2, Garmin Vivosmart, Apple Watch, and Xiaomi Mi Band. The findings claim that all tested devices, except Apple Watch, use the static device addresses that allowed attackers to track user information such as location, time of fitness activities, and reversing user profile by eavesdropping on these devices' communications. Ryan [167] described how to perform eavesdropping on a BLE device communication by using the Ubertooth sniffer tool. This work also demonstrated how to successfully bypass the encryption of BLE wearables that use BLE legacy pairing by breaking the encryption key.

b) Traffic analysis attacks: In the context of wearables, traffic analysis attacks are the processes of monitoring traffic exchanged between wearables and its base and/or server and make inferences from patterns of the communication. For instance, Das *et al.* [171] discovered privacy leakages in relation to the Bluetooth LE communication between the fitness tracker and the smartphone including *User Tracking*, *User Activity Detection*, and *Person Identification*. The authors observed that the fitness trackers constantly advertise even when they are in close proximity of the smartphone. By sniffing on the BLE advertisements and leveraging the unchanged device address, adversaries are able to map a specific device to a specific user. Consequently, the consistent advertising and the use of static address in BLE fitness trackers make them vulnerable to *User Tracking*. The authors also discovered the correlation between the BLE data traffic of a Fitbit and a

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smartphone with the intensity of user's activity, as the BLE traffic consists of data packets and empty packets (without payload) when there is no meaningful content. Based on the size of the packet being transferred and the number of packets in the BLE channel, an adversary can simply sniff the BLE traffic and infer the user's activities even though the data packets are encrypted. Another finding in this work is that the BLE data can represent the unique walking pattern of a user, making it possible to identify a person from a small group of people even when random addresses are used.

Recently, Lotfy and Hale [172] observed the pairing and data exchange mechanism in three wearable products, Jawbone UP, Fitbit Charge HR, and Pebble Steel Watch, that each uses different pairing models. The tested setting consists of an *Ubertooth One* sniffer [176] connected to a Linux-based machine for eavesdropping BLE packets, an iPhone and a wearable device. The experiment investigated the ability of an attacker to intercept the advertising, pairing, and data packets sent from the wearable to the iPhone. The authors reported that they were able to detect connection request packet and observe subsequent data exchanges in all three wearable devices.

c) Information gathering attacks: As mentioned previously under *Eavesdropping Attacks* and *Traffic Analysis Attacks*, passive monitoring of data transmissions on wearable devices allow adversaries to collect information exchanged between a wearable and its hub. Consequently, such collected information can be used to conduct serious *information gathering attacks* such as breaking the key exchange in BLE wearable device pairing or to collect information about user's other devices such as smartphones.

Ryan [167] demonstrated an attack that can break the *LE legacy pairing* of BLE communications of wearables. This work exploits the fact that in *LE Legacy pairing*, all pairing related information values other than a secret Temporary Key (TK) are transmitted over the air in plaintext. The TK is used to generate a Short Term Key (STK) at both devices that is then used to encrypt the link between them. Then, the encryption key called Long Term Key (LTK) is generated and distributed between two devices for future connections over the encrypted link by STK. For breaking the encryption key, Ubertooth tool is used to eavesdrop on the plaintext transmitted values. An offline brute-force attack is then used to guess the TK. Once TK is cracked, the STK is easily recovered. Consequently, the LTK will be cracked and attackers are able to decrypt future conversations between BLE devices. Therefore, wearables used BLE legacy pairing are prone to the LTK cracking attack that relies on the eavesdropping and brute-forcing.

Maiti *et al.* [173] investigated the vulnerabilities of using wearables such as smart watches for side-channel attacks. This work demonstrated that leveraging the motion sensors in smart watches as side-channel to infer sensitive user information is even better than motion sensors in the smartphone. Specifically, the adversary can make use of the wrist movements perceptible by a target's smartwatch motion sensors to infer the target's keystrokes on a smartphone numeric touchpad. Adversary gains access to the smartwatch by installing a malicious application on it to record activities of the on-board accelerometer and gyroscope sensors.

Similarly, Liu *et al.* [174] demonstrated a practical side-channel attack to infer user inputs on the keyboards by exploiting sensors in smart watches. By extracting the motion data from the accelerometer in the smart watch and modelling the captured movements using displacement or motion directions, the authors can decode the keystrokes in a numeric keypad of a Point of Sale (POS) device and a normal QWERTY keyboard.

Recently, Wang *et al.* [175] demonstrated a practical attack to infer a user's personal PIN sequence by exploiting her wearable devices including smart watches and fitness trackers, when he is accessing a key-based security system. In particular, the attack model exploits the unique patterns of key entry activities from the sensor data captured by wearables to develop distance estimation and slope-based moving direction derivation schemes that captures the limited movements of the hand between two consecutive key presses. Furthermore, the attack model applies the backward PIN-sequence inference algorithm to reveal the user's complete PIN sequence, leveraging both the spatial and temporal constraints of the key entry. This methodology is reported to achieve a high success rate. To extract the sensor readings of wearables, adversaries can simply sniff Bluetooth LE communications between the wearables and their paired smartphone or install malicious app on the wearables.

d) Other attacks: In addition to three common threats to confidentiality discussed above, there are a limited number of work targeting the vulnerabilities of wearable systems or the associated applications. For example, Goyal *et al.* [169] analysed the security aspects of the Android apps of Jawbone and Fitbit. Authors recreate a scenario where a smartphone is infected by a malicious application that can access the received raw data from the tracker. The finding is that the Jawbone application sends main information about the device to third parties for different processing capabilities. In both Jawbone and Fitbit devices, the application preferences and data are stored unencrypted on the device. The local databases storing all user activities are in plaintext allowing access to any attacker. While Fitbit allows users to choose which data will be shared, the users of Jawbone UP Move tracker do not have any control or ownership over their data.

A recent study carried out by HP [178] showed that all tested consumer smartwatches lacked some form of critical security measures such as data encryption or user authentication. For example, it was found that 50% of the devices did not have a basic screen lock implemented and two devices could be paired with an attacker's phone in a stolen scenario. Moreover, 70% of the devices push their software updates unencrypted.

To summarise, majority of the reported vulnerabilities are due to the inadequate implementation of security measures by the devices manufactures rather than fundamental security limitations in Bluetooth LE. One possible reason behind this is that manufactures are trying to launch wearable products quickly to get established in the market and as such security is not their primary focus at this stage. Also, another factor can be the trade-off between the constrained resource of

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wearables and the complexity of security measurements. As happened in the smartphone domain the next generation of the consumer wearable products are likely to address these security vulnerabilities. However, the resource against security trade-off will still be there and effective countermeasures to protect the confidentiality in wearable communication is essential.

2) Threats to Integrity: Integrity is an important security requirement for information systems especially for wearable systems where collected data is usually sensitive and private. It is important to make sure that data is not altered in transit and being received by authorised parties only. Several work in the literature have assessed wearable device systems with respect to data integrity and highlighted potential vulnerabilities and we discuss such work below under three attack categories; *i) Modification Attacks*, *ii) Replay Attacks*, and *iii) Masquerade Attacks*.

a) Modification attacks: As data transmission between wearable devices is typically wireless on the air, the data is prone to be modified or altered. In these types of attacks, the adversaries after either intercepting the traffic exchange of wearable devices or gaining access to the information, can modify the content of exchanging packets or change the timestamp of data packets.

Haataja and Toivanen [179] demonstrated the *Just Works* model in Bluetooth LE pairing makes wearable devices vulnerable to modification attacks. Specifically, the information exchanged between wearables and their base is modified, i.e., the input/output capabilities information in the pairing packet exchange is changed to "No Input No Output", so that the victim wearables choose to use the Just Works pairing mode allowing the attacker to intercept and modify subsequent data exchanged between two victim devices.

Rahman *et al.* [180] found vulnerabilities in personal fitness data storage and transmission in popular trackers from FitBit and Garmin which use the ANT protocol for the communication between a tracker and its base - a smartphone. This work identified several fundamental vulnerabilities and then simulated attacks to exploit these vulnerabilities. First, the Fitbit tracker does not use encryption during user login via the client software which means user passwords are sent to the Web server in plaintext. Second, both FitBit and Garmin do not use encryption or authentication whilst uploading data. Third, there is no mutual authentication process during the pairing between Garmin tracker and its base, i.e., the base authenticates the tracker but the tracker does not authenticate the base. The authors built two attack tools called *Fitbite* and *GarMax* to exploit the identified vulnerabilities in Fitbit Ultra and Garmin Forerunner trackers. Results showed that FitBite can discover tracker devices within transmission range and capture their fitness data during data uploads. GarMax can directly download fitness information from the tracker as the Garmin tracker does not authenticate the base. FitBite and GarMax are able to modify and inject fitness data on neighbouring trackers.

Recently, integrity of the timestamps in sensor data from wearable healthcare devices is discussed in [181]. The correct timestamps of medical data is crucial for medical practitioners and insurance providers and attackers/malicious users might attempt to modify the timestamps to obtain monetary benefits. The authors demonstrated that they can tamper the timestamps of measurements from a smart blood pressure monitor called Fora Diamond Cuff BP, by simply changing the time on the paired smartphone clock while taking the blood pressure. A similar attack was also performed on an activity tracker, Withings Pulse O2. Moreover, another timestamp tampering attack was done by using the Apktool to decompile the Android apps supporting the Fora and Withings blood pressure monitor devices and making modifications to the source code, and recompiling code in an Android smartphone. The source code modifications will make sure that the timestamps of medical data will be tampered when they are being uploaded to the smartphone.

The Open Effect Report [170] discovered that while most fitness device applications used HTTPS to encrypt their communication with the servers the Garmin Vivosmart's application does not provide HTTPS transmission and exposed the sensitive fitness data to the unauthorized third parties. This work identified that the fitness data generated by several wearable devices can be falsified by motivated parties.

b) Replay attacks: In replay attacks, attackers obtain a copy of the valid data packet uploaded by the wearable device to the server and replay it with the objective such as impersonation or data corruption. The only work we found in this category related to wearables was by Li *et al.* [182] where authors demonstrated a possible *replay attack* in a commercially available insulin delivery system consisting of a glucose meter, an insulin pump, a remote control and a Universal Software Radio Peripheral (USRP). The communication frequency between the remote control and insulin pump is 915 MHz. The USRP can receive and generate signal in the same frequency band through an attached 915 MHz daughter board and an antenna. By eavesdropping on the communication between devices, attackers are able to determine the packet format and learn the device type, its PIN, the therapy or glucose level, and medical condition of the patient as they are transmitted in plaintext. In addition, by using brute-force method it is possible to figure out the Cyclic Redundant Check parameters used in the insulin delivery system. Then, by using a different value in the *counter* field of the packet, *replay attack* was explored for reporting an outdated glucose level.

c) Masquerade attacks: In masquerade attacks the attacker impersonates or clones an authenticated device to act as the legitimate node for stealing data or injecting fake information to the system.

Naveed *et al.* [183] identified threats of "external device mis-bonding" in Bluetooth-enabled Android phones. In fact, any apps with the Bluetooth permission running on the authorised Android phone can communicate with the external device which is paired with the phone. Due to the lack of a secure bonding between healthcare devices and their corresponding apps in the paired phone, the malicious app running on the Android phone collects the bonding information from the medical device and delivers it to the adversary. The adversary then *clones* the legitimate device and places the clone in the proximity of this device. It is possible to inject fake data into the

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device's official app or user's online account when the user is close to the clone's Bluetooth device and the malicious app unpairs the phone from the original device and pairs it with the clone. In addition, as the malicious app can get access to patient's health data through the Bluetooth interface of the healthcare device, it can steal sensitive data by monitoring when the target apps launches, and connecting to the external device in a small time window.

Previously mentioned work by Li *et al.* [182] (***see Replay Attacks***) also identified that once an adversary knows the device's PIN of the remote control of the insulin delivery system, he can use the Universal Software Radio Peripheral to control the insulin pump by generating unauthenticated control commands, for example, to stop/resume insulin injection into the human body.

In summary, the attacks against data integrity in wearable systems are accomplished due to the lack of authentication methods and/or the unencrypted data exchanged over the air between wearable devices. We noticed as of now threats to integrity are not as commonplace as threats to confidentiality. Nonetheless, several work highlighted some possible attacks in medical wearable devices for which the data integrity is much more critical compared to consumer wearables. Majority of the attacks are possible due to the vulnerabilities in data confidentiality and solutions for those will also alleviate threats to data integrity.

3) Threats to Availability: The major type of attack against availability is *Denial of Service* attacks that attempt to interrupt the communications of wearables and their base or inject huge useless information to flood the wearable device's storage capacity. We found two such types of example attacks in the existing body of research literature.

Goyal *et al.* [169] discovered that the *FitBit Charge* tracker is susceptible to Denial of Service (DoS) attacks which can keep the tracker always busy by making it respond to every request to prevent the legitimate device syncing with the paired tracker or to prevent the unpaired tracker from pairing with the mobile phone. Previously reported attack tools (***see Modification Attacks***) [180] Fitbite and GarMax [180] were able to prevent the Fitbit and Garmin trackers from recording the user's valid data by injecting a huge amount of fake data exceeding the storage capacity of the trackers. In addition, the FitBite attack tool could simulate attack scenarios where adversaries continuously query the trackers in its vicinity to drain their batteries.

Similar to the other threats we explained previously, current attacks denying service is possible only because of implementation shortcomings by the manufactures and can be expected to be resolved in next generation of products. Nonetheless, it is essential to continuously look for such vulnerabilities in wearable products, especially the health related products such as insulin pumps which can't accomodate downtimes.

B. Proposed Security Solutions for Wearables

Having explored the existing threats and vulnerabilities in current consumer wearables, in this sub section we survey solutions proposed by the research community to address threats towards communication security in wearables. We present the solutions according to the type of security threat they are attempting to address. There were many solutions that target threats to both confidentiality and integrity and we therefore present them in one section.

1) Ensuring Confidentiality and Integrity:

a) Key generation and exchange: Encryption is the *de-facto* method of ensuring confidentiality in communication over a network. All the commonly used wireless protocols in wearables offers some form of MAC layer encryption. However, as we highlighted in Section III-A, MAC layer encryption might not be sufficient in some cases. As a result, number of work focused on supporting higher layer encryption protocols. However, when it comes to wearables, its resource constraints such as CPU, memory, and energy do not allow to implement advanced encryption methods. For example, Snader *et al.* [184] measured that in order to encrypt a block of data using ECDH-ECDSA key exchange, requires 6,000 times higher energy compared to symmetric encryption. Differences in such magnitude is crucial in wearables due to the limited battery power. Also, the limited payload size in some protocols such as Bluetooth LE, does not allow to transfer the longer keys in single message and thus key exchange can be disrupted significantly under unreliable channel conditions. To this end, number of solutions were proposed for *low power encryption protocols* and *key exchange mechanisms* on wearables.

Several studies focused on mechanisms that allow two wearables to agree on a symmetric key without the need of transmitting the complete key over the wireless channel [184]–[187]. Venkatasubramanian *et al.* [185] proposed Physiological-Signal-based Key Agreement (PSKA) for secure inter-sensor communication for body area networks. Nonetheless, PSKA is also applicable for wearables. In PSKA two sensor nodes agree to a symmetric (shared) cryptographic key, in an authenticated manner, using physiological signals obtained from the wearer. More specifically, first the *sender* generates a random symmetric key which will be hidden using a feature vector generated from the physiological signal. This hidden key is then sent to the *receiver* who will be able to recover the random key using its own version of feature vector (generated by the receiver using its own measurements of the physiological signal). Main challenge to address here is how to compensate the difference between the feature vectors and the measurements taken from the two nodes can be different according to their placement. To solve this, authors used the *fuzzy vault* technique in cryptography. *Fuzzy vault* allows to lock (hide) a secret (S) in a construct called a vault using a set of values A. Once the vault has been locked, it can be unlocked only with another set of values B that has a significant number of values in common with set A and that suits the scenario where physiological signal is measured from two nodes. Revadigar *et al.* [186] used a similar approach with the accelerometer signal.

Hu *et al.* [188] proposed Ordered-Physiological-Feature-based Key Agreement (OPFKA) that also follow a same approach as fuzzy vault techniques. First, the sender and the receiver collect Inter Pulse Intervals (IPIs) simultaneously to

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obtain a fixed amount of measurements. The sender then conceals his measurements in a *Coffer* by creating a set of chaff points and permuting them with the measured signal values. The coffer is then passed to the receiver which can identify the indices of the actual measured values than chaff points by comparing with its own measurements. These values are then used to create the shared key. Authors compared their solution with PSKA [185] and showed that OPFKA is more secure at a lower computational cost.

Fuzzy vault techniques originally comes from biometric authentication, and some work have demonstrated that an attacker may be able to guess the legitimate points in the vault [188]–[190]. However, compared to fingerprints or any similar form of static biometrics, signals such as pulse rate and accelerometer are more dynamic and noisier and as a result it expected to be difficult for an attacker to identify the legitimate points.

An alternative approach to fuzzy vault based key establishment is *key generation* [191]–[193]. These methods directly create the keys rather than relying on vault based techniques to conceal the key and exchange. For example, Shi *et al.* [191] proposed ASK-BAN, where key extraction is done based on measurements of the wireless channel characteristics between a controller unit (CU) and an on-body sensor node. First, device authentication ensures the sensor devices that communicates with the CU are on the same body of one person, using the stable channels between on-body devices compared to the off-body devices that have highly unstable channels to the CU due to simple body movements that the patient artificially performed during the process. Simultaneously, secret keys are extracted between every authenticated on-body device and the CU, utilising their unstable channels.

Another approach to key distribution is the use of *out of band channels*. For instance, Snader *et al.* [184] proposed *CryptoCop* encryption protocol for wearable mobile devices. The main idea behind *CryptoCop* is to utilise the charging times of the wearables (e.g., via USB cable by connecting to a laptop) to upload the key information in comparison to a conventional key exchange happening over the wireless channel. To address the issues of packet size and unreliable channel conditions, *CryptoCop* uses AES in counter mode in contrast to CBC (Cipher Block Chaining) and CFB (Cipher Feed Back) modes used in classical protocols. The performance of the protocol is evaluated using wearable heart rate monitors.

Another out of band key exchange mechanism *SecureVibe* by Kim *et al.* [194] includes an ultra-low power battery drain resistant wakeup mechanism and a vibration-based, lightweight cryptographic key exchange scheme between the wearable medical device and the external device by exploiting the advantages of vibration channel such as extremely short range, direct physical contact requiring and highly user-perceptible. Specifically, the authors proposed an enhanced two-feature On-Off Keying scheme for vibration channel to enable acceptable bit rates for key exchange. In the wakeup mechanism, the vibration signal is detected through the vibration measurement of an accelerometer in the wearable medical device. A high-pass filter is used to eliminate the low-frequency noise from patient movement or internal organs. The RF module is only waked up if a high-frequency vibration signal is detected after the filtering. After waking up the RF module in wearable device, a shared symmetric encryption key is exchanged between the wearable and the external device. In the key exchange scheme, the external device generates a key w and modulates it into vibration signal. Upon receiving the vibration signal, the wearable uses two-feature OOK demodulation scheme to convert signal to a bit string w'. The wearable then encrypts a predefined, fixed confirmation message using w' and transmits it to the external device through RF channel. The successful decryption of the confirmation message using key w at the external device implicates the successful key exchange and subsequent communication is encrypted by using the key w.

Kim *et al.* [195] proposed an encryption module to enhance secure data transmission between wearable devices with lightweight micro-controller MSP430. The Advance Encryption Scheme (AES) with optimal code size is implemented and compiled using "msp-gcc" for MSP430. In the designed wearable device's system architecture, the encryption module with two interfaces for setting secret key and encrypting a cipher block is placed between the application layer and transmission layer. As a result, all healthcare applications can use this service before transmitting data to the cloud server.

To summarise, low power *Key Generation and Exchange* schemes usually attempt to facilitate a key agreement between the sender and receiver without actually transmitting the key in the wireless channel using some common data both the sender and receiver has access to, e.g., vital signs of the wear or using another secure channel to transfer the key. Nonetheless, majority of the above work operate under the premise that wearable devices are not capable enough to do asymmetric encryption due to their resource constraints and as a result secure key sharing methods are required for symmetric encryption key sharing. More recent work [196] have showed that some state-of-the-art wearables are capable of asymmetric encryption without significant overheads. We believe this trend might continue in some wearable device types such as smart watches and fitness bands. As a result, low power key generation and exchange procedures are likely to be used by more compact devices such as smart ear-buds.

b) Authentication protocols: Authentication protocols allow securely transfer authenticated data between two entities. Such protocols are essential as encryption alone does not necessarily guarantee integretiy in communication where malicious nodes can impersonate servers or relay units of wearables to obtain unauthorised access to data.

Rahman *et al.* [197] proposed *FitLock*, a lightweight defence system to protect Fitbit. *FitLock* implements a *bind* procedure which allows the user to bind his new tracker to his social network account, and an *upload* procedure which assures the security for the Fitbit communication protocol. During the bind procedure, the *BindTrackerUser* protocol between the user, tracker and the Fitbit Web Server (WS) is conducted that takes user account ID, tracker ID, shared secret symmetric encryption key between the tracker and the WS, and database Map structure of the WS as input arguments. After several steps exchanging these arguments, the user enters a

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6-digit random value N created by the WS into a confirmation box in his Fitbit social network account. If N is confirmed by the WS, the tracker is binded with user's social network account. Moreover, the *UploadData* protocol in upload procedure is conducted to secure the communication between the tracker and the WS. The secret key is used to encrypt all communications between them and to authenticate the WS once the tracker has received a request from it. *FitLock* also verifies the consistency of the data reported by trackers. For instance, it utilises the walk/run stride length and the user "Basal Metabolic Rate" values to verify the relations between the number of steps and the distance traversed and the calories burned by the user.

In a following work, Rahman *et al.* [180] proposed *SensCript*, a secure protocol for data storage and transmission in fitness centric social networks with lightweight personal trackers. It is used not only to thwart attacks such as tracker private data capture, tracker injection, and user account injection, but also to defend against attackers that physically capture and read the memory of trackers. More specifically, tracker (T) shares a symmetric key K_T with WS while WS also maintains a unique secret key K_W for each tracker. All transmissions between T and WS are authenticated with K_T. In addition, each data record stored on tracker is encoded using two pseudo-random numbers generated by the tracker using K_T when the record is written on tracker's memory, and by WS using K_W and written on tracker during data sync procedure.

To provide a new end-to-end authentication for wearables, the work in [198] integrated a Java Card Technology-based smart card (JCard) into the wearable device (WD) providing the wearable with the ability of NFC connectivity and self-authentication, which is called WD-NFC-JCard entity. The proposed authentication protocol defines two different authentication flows. One is authentication flow is between the intermediate device and WD-NFC-JCard whereby the user registers a trusted immediate device with the WD-NFC-JCard. Then a NFC communication channel is set up between the WD-NFC-JCard and immediate device to exchange and verify public key certificates and digital signature. The other authentication flow is mutual authentication between the WD-NFC-JCard and the remote server assuming that they are aware of and trust the public key of each other. In this mutual authentication, a trusted immediate device is used to forward massages exchanged between them.

In order to protect Bluetooth-enabled health/medical Android devices from the *external Device Mis-Bonding*, Naveed *et al.* [183] proposed *Dabinder*, an OS-level access control that automatically generates and enforces secure bonding policies between device and its authorised apps. *Dabinder* controls app-device bonding in both pairing procedure and communication stages. More specifically, once a device is activated, the user performs the pairing procedure between the device and its authorised apps whereby *Dabinder* generates a bonding policy which associates each device to its official apps. When an app attempts to establish a Bluetooth connection with a device, a policy enforcement mechanism checks if the app is associated in the bonding policy to the device. If the app is in the device's bonding policy, the request is allowed to proceed, otherwise, it is declined. In addition, *Dabinder* performs an un-pairing controller to manage pairing relation between the phone and a device. Specifically, when an app tries to un-pair the phone with a device, un-pairing controller will find whether a bonding policy between the device and this app is available. If there is no bonding policy found, the app is reported as unauthorized and its un-pairing attempt is stopped.

Authentication Protocols majorly attempt to address the challenges related to data integrity and ensures that the sender is communicating with the right receiver. The solutions we discussed in this category come together with a encryption solution and no wearable related work is found to tackle the authentication challenges alone. Apart from software based solutions, we found that there are hardware based solutions tailored for NFC-enabled devices, that might find applications in smart jewellery.

2) Ensuring Availability: As described in Section III-A, threats to the availability of wearables are limited and as a result not many work focused on that aspect of wearable security. In [197], the proposed solution *FitLock* not only protects Fitbit from data capturing and data injection attacks but also thwarts the DoS and Battery Drain attacks. As all communications between the tracker and Web server are encrypted with their shared key and each communication session has a monotonically increasing session ID, *FitLock* prevents attackers from obtaining data in uploading data process or replaying the requests. The tracker simply drops the invalid requests or requests with old session IDs, counters. As a result, it does not consume resource for processing those requests.

C. Summary of Communication Security

In this section we surveyed the work that identified communication security threats in wearables and highlighted proposed solutions to those threats. Several work highlighted the issues of wearable device's communication in terms of confidentiality, integrity, and availability caused by improper implementations of communication technology such as lack of encryption, insufficient authentication methods, and not using MAC address randomisation. Majority of the work highlighted vulnerabilities threatening confidentiality and integrity. The proposed solutions predominantly focused on two aspects; *key generation and exchange procedures* and *authentication protocols* that address threats to both confidentiality and integrity.

We also noticed that existing work are mainly focusing on investigating communication security threats in BLE enabled wearables as it is the most popular choice of communication among consumer wearables. There were some limited work in vulnerability analysis on NFC and ANT/ANT+ and we believe further security analysis is required in these protocols as limited footprint devices such as smart jewellery are likely to be using these protocols in future when they become mainstream.

IV. Energy Efficiency

Energy is considered one of the most critical resources in wearable computing, especially with increasing user demands

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TABLE II
WEARABLE BATTERY TYPES

Type	Characteristics	Devices	Limitations
Lithium Coin	Small in size, lightweight, inexpensive, and fairly safe.	Small and low-power wearables.	Non-rechargeable and have low current draw capability.
Lithium-ion (Li-ion)	Rechargeable, lightweight, have higher power density and cheaper in cost than Li-Poly, do not require priming when first used and have a low self-discharge.	Many wearables including wrist-worn and head-worn.	Suffer from aging even when not in use.
Lithium Polymer (Li-Poly)	Rechargeable, lightweight and flexible (more than Li-Ion) and have improved safety.	Many wearables including wrist-worn and head-worn.	Cost more to manufacture and have a worse energy density than Li-ion batteries.
Graphene	Promised to be more efficient, more flexible, have higher energy density and storage capacity than all of the current battery types.	Still under development.	Expected to be more expensive than other batteries.

for more power and advanced functionalities. It is not convenient for many users to frequently recharge their wearable devices by plugging the devices into to power sources. More recently, wireless charging is able to offer more flexibility and convenience to the users. However, it usually needs longer charging times since the rate energy can be transferred to the device is limited [199]. Thus, as highlighted in Section II, active battery life is a critical factor the users consider when it comes to selecting wearable devices. This has motivated both the industry and research community to explore for means of extending the battery lifetime of wearable devices.

Energy efficiency of wearables can be generally achieved by either advancing the technology of batteries or minimising the consumption of power. Recently, energy harvesting [200] is also receiving attention of both research and industrial communities. In this section, we enumerate the work related to energy efficiency in wearable mobile devices under three topics. First, under *Battery Advancements* we describe the state-of-the-art in battery technologies, their limitations, and future trends. Next, under *Energy Efficient Sensing* we describe methods that is proposed to minimise the sensing power consumption without compromising the system performance. Finally, we describe research in *Energy Harvesting* in related to wearables.

A. Battery Advancements

Batteries are currently a major design factor in wearable devices. To maintain the growth of the wearable market, batteries used for wearables should be compact, efficient, and long lasting. In the last few years, we have witnessed an improvement in the batteries technology trying to meet the wearable constraints by considering the battery size, shape, capacity, and reliability [201].

There are different kinds of batteries which are used in present day wearables [201], [202]. Lithium is still the key element in wearables batteries because it is the lightest of all metals and it operates at a high voltage [202]. Lithium coin or button cells batteries are one of the widely used batteries for small devices such as remote controls, hearing aids, watches, and some of the earliest wearable devices [201]. These types of batteries are well-suited to small and low-power wearables as they are compact, light-weight, cost-efficient, and fairly safe. However, they are not rechargeable and have high internal resistance [203].

On the other hand, Lithium-ion (Li-ion) and Lithium Polymer (Li-Poly) batteries are becoming the most common in many wearable devices including smartwatches, wristbands and head-mounted wearables as discussed in Section II. Both the Li-Poly and the Li-ion batteries are rechargeable, lightweight, and have high power density. However, they require special circuitry to keep them from exploding. They usually come with protection circuitry that ensures the safe operation of the battery [204]. The Li-Poly battery (also called Lithium-ion polymer or a pouch cell design battery) is lighter, more flexible, and have improved safety than the Li-ion battery. On the other hand, the Li-ion batteries are cheaper in cost and have a better energy density than the Li-Poly [205] but they generally suffer from aging even when not in use.

Recently, graphene-based batteries have emerged as an alternative. They are promised to be more efficient and flexible with higher energy density and storage capacity than all of the current battery types and is expected to advance the wearable technology. However, they are still under development and expected to be much more expensive than other batteries [206]. Table II shows a summary of wearable battery types including their characteristics, the types of wearables they are suitable for, and their limitations.

Although, there is an extensive work going on improving batteries technology to meet the wearable constraints, batteries are still one of the largest limitations in advancing the wearable technology. Battery technologies have not evolved in the pace the wearable products are advancing with respect to the hardware/software features and as a result, research community has looked into the possibility of minimising the power consumption of wearables. We next discuss the available research studies which achieve energy efficiency through minimising power consumption.

B. Energy Efficient Sensing

Sensing can be highly power consuming as a result of using many sensors in the wearable and the continuous sampling of such sensors. Minimising sensing power consumption can be

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achieved by turning off power hungry sensors when ever possible, using the minimum number of sensors to achieve a desired application, and reducing the sampling rate per each sensor. However, this can degrade the performance of the system. Therefore, minimising sensing power consumption without affecting the system performance is an important factor to be considered while achieving energy efficiency. In this subsection, we review some of the proposed research studies to reduce sensing power consumption in wearable devices.

1) Sensor Selection: Sensor Selection is a method to minimise sensing power consumption by selecting subset of sensors to be turned on and turning off other sensors to save power. For example, turning off power hungry sensors such as GPS and WiFi can significantly minimise sensing power consumption. Consequently, some research studies investigated the use of low-power inertial sensors such as accelerometers and gyroscopes instead of the power hungry GPS sensor for pedestrian tracking applications [207], [208]. Inertial sensors are known to consume much less energy than GPS. However, they can add various challenges when used for tracking, such as different phone placements and phone orientation which might affect the performance of the system. To address these challenges, Zhu *et al.* [207] introduced a robust dead reckoning algorithm and an error-tolerant map matching algorithm which achieves outdoor pedestrian tracking using low power inertial sensors without affecting the performance of the system. Similarly, Abdesslem *et al.* [208] proposed a system for indoor/outdoor pedestrian tracking which can use the low-power accelerometer sensor more often and the more expensive GPS sensor less frequently. They relied on the intuition that when the user is not moving, using sensors for localisation is unnecessary. The accelerometer can thus be used to detect when the user is not moving and then stop GPS sensing to save energy. They showed that their system can reduce sensing energy consumption by more than 58% for localising a user indoor and outdoor.

Although inertial sensors are known to be energy efficient compared to GPS and WIFI, the continuous sampling of many inertial sensors included in the wearable can be also power consuming. Zappi *et al.* [209] introduced a sensor selection scheme to reduce inertial sensing power consumption for human activity recognition application. The system exploits the redundant and unreliable accelerometer sensors and relies on a meta-classifier that fuses the information of classifiers operating on individual sensors. Sensors are selected according to their contribution to classification accuracy as assessed during system training. Their results showed that the lifetime of the system can be significantly extended while keeping high recognition accuracies.

On the other hand, Wang *et al.* [210] proposed an energy efficient mobile sensing system (EEMSS) which not only recognise user activities/states but also state transitions. They incorporated a hierarchical sensor management scheme which selectively turns on the minimum set of sensors to monitor user state and triggers new set of sensors if necessary to achieve state transition detection. By powering only a minimum set of sensors and using appropriate sensor duty cycles, the authors showed significant improvement in device battery life (more than 75%) while maintaining both high accuracy and low latency in identifying transitions between end-user activities. Moreover, Grordon *et al.* [211] proposes a sensors selection method for activity recognition which leverages the predictability of human behaviour to conserve energy. By predicting human behaviour, sensors which are not needed to recognise the probable activities from each other can be turned off conserving energy. Their results showed significant energy savings of up to 84% without greatly impacting recognition rates.

2) Adaptive Sampling: Adaptive sampling is a widely used method to reduce sensing power consumption by reducing the sampling rate of sensors. However, this reduction should be achieved without affecting the information precision and the system performance. Krause *et al.* [212] analysed the trade-off between reducing the accelerometer sampling rate of *eWatch* wearable device and the system performance in human activity recognition application. They used a collection of selective sampling strategies in order to reduce the number of required sensor readings. Their results indicate the existence of a sampling threshold, below which accuracy falls dramatically. By selecting this sampling threshold, the lifetime of the device can be extended by a factor of four without a significant loss in activity recognition accuracy.

On the other hand, Yan *et al.* [213] studied the tradeoff between reducing the accelerometer sampling rate and human activity recognition accuracy *separately for each activity*. They showed that such tradeoff is activity specific which means different activities require different sampling rates to achieve the desired accuracy levels. Based on that finding they introduced an activity-sensitive strategy, called A3R, which adapts the sampling frequency in real time, based on the activity type. They showed that for users running the A3R application on their Android phones an overall energy savings of 20-25% could be achieved. Following this idea, Qi *et al.* [214] proposed AdaSense algorithm which uses a lower power single activity event detection most of the time. It only resorts to multi-activity classification when an activity change is detected. Using Genetic Programming approach, the authors further reduced the sampling rates of both single activity event detection and multi-activity classification, leading to 39.4%-51.0% smartphone energy savings compared to the A3R application.

While it is possible to extend the battery lifetime of wearables through the previously mentioned energy-efficient sensing solutions, battery-powered wearables cannot provide sustained operation without the need for battery recharge and eventually replacement. This motivates many researchers to explore energy harvesting solutions to provide self-powered wearables. In the next subsection, we discuss the available energy harvesting-based solutions.

C. Energy Harvesting

Building self-powered devices by harvesting energy from the ambient environment is extremely attractive for wearable computing [215]. Instead of relying on batteries to power the wearables, energy harvesting (EH)-enabled wearables can

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freely gather energy from ambient sources, such as solar, kinetic, and heat, to supply or recharge the electronic directly without the needs of manually battery recharge or replacement.

In this section, we provide a relatively high-level review of existing works that investigate harvesting energy from the ambient environment to build *self-powered wearable devices*. For seeking more details about the fundamental mechanism of different energy harvesting techniques, and their usages in other disciplines, we encourage readers refer to [216] and [217].

In the rest of this section, we introduce recent advancements and applications of energy harvesting in wearable devices. Depending on the source where the ambient energy is harvested from, current EH-based solutions can be classified as; **1) Kinetic Energy Harvesting**, **2) Thermoelectric Energy Harvesting**, and **3) Solar Energy Harvesting**.

1) Kinetic Energy Harvesting (KEH): KEH has emerged as a viable option to power the wearables by converting the kinetic energy of human motion into electrical energy. Several research studies started to build practical wearable devices powered by human motion in different form factors [218]–[220].

Shoes is considered the most popular form factor in designing KEH-powered wearables. The history of building shoe-based self-powered wearable can back to the late nineties, during which the researchers in MIT started to integrate magnetic generator with shoes [221]. In an earlier work of Antaki *et al.* [222], the authors discovered that the ground reaction forces associated with the heel strike and toe-off phases of the gait can generate the largest amount of energy during human walking. In this study, a piezoelectric array-based EH shoe has been built to generate electric energy from human gait. Similarly, in the work of Kymissis *et al.* [223], a piezoelectric generator is placed inside the shoe and can generate 1.1-1.8mW average power during walking. They have proved that the generated energy is able to power the RFID transmitter to broadcast signal periodically. In addition to piezoelectric based solutions, in the work of Duffy and Carroll [224], an electromagnetic generator was used inside the shoe and an RMS voltage of 500mV could be generated by the prototype at a 5 Hz frequency.

In a more recent study conducted by Huang *et al.* [225], a prototype of self-powered shoe is presented. The prototype consisted of an accelerometer and Bluetooth wireless communication unit which are powered separately by the energy harvested from each of the two feet, and coordinated by ambient backscatter. Such that, the accelerometer can sense the activity of the user, while the Bluetooth can transmit the sensing results to the smartphone.

Many works in KEH-powered wearables also built their prototypes in form factors other than shoes. As reported in the works of Granstrom *et al.* [226] and Feenstra *et al.* [227], a novel energy harvesting backpack has been built, which can effectively generate electrical energy through the differential forces between the wearer and backpack. Their experiment results shown that, in average, approximately 0.4mW of power can be generated using their prototype, which is certainly sufficient to power some electronics with low energy consumption. In the work of Yang and Yun [228], a fabric wearable platform embedded with piezoelectric energy harvester has been designed, which can be worn on the elbow joints and fingers. Their experiments shows an average output of 0.21mW can be generated by the fabric from the movements of elbow joints and fingers.

2) Thermoelectric Energy Harvesting (TEH): Thermoelectric energy harvesting (TEH) converts the temperature differences between the ambient environment and human body into electrical energy to power wearable devices.

The most popular TEH-powered wearable device is in the form-factor of watch. In 1980, Bulova introduced the prototype of Thermatron [229]. Thermatron harvests energy from the temperature difference between the back of the watch, which contacts with the wrist of the wearer, and the top of watch, which contacts with ambient air, to power the quartz caliber. A similar example of TEH-powered watch is the Seiko Thermic wristwatch [230], which also uses the temperature gradient between wrist skin and atmosphere to generate energy. In the work of Leonov *et al.* [231]. A watch-size thermoelectric wrist generator was fabricated, and can generate approximately 250μW in average during the daytime, using the temperature difference between human skin and ambient environment. They further proved the practical use of their prototype by using it to power a wireless pulse oximeter (S_PO_2 sensor) [232], [233].

As reported, the power consumption of the oximeter is typically 62μW, while their thermoelectric wristwatch can generate approximately 200μW during the daytime, which is sufficient to power the whole system. Following the success of this work, they developed a battery-free body-powered electroencephalography (EEG) to monitor electrical activity of human brain [234], and electrocardiography (ECG) shirt to record the electrical activity of heart [235].

3) Solar Energy Harvesting (SEH): Because of the abundant energy source and high power density, energy harvesting from solar and ambient light has been widely studied in powering and extending the operational time of wireless sensor networks [236]–[238]. In generally, solar energy harvesting can achieve a power density up to 100mW/cm^2 under outdoor sunlight and 100μW/cm^2 in indoor environment [215], which is sufficient to power many wearable systems with low-energy consumption.

Similarly, wrist watch is the most widely used form factor for SEH-powered wearable devices. Many manufacturers have released their own SEH-powered wrist watch [239]–[241]. For instance, the SEIKO has launched the first Solar watch [239] in 1977. The watch utilises a solar cell to harvest energy from ambient light and power an rechargeable battery. With 1 minute quick harvesting from sunlight, it enables up to one hour regular operation of the watch. Misra *et al.* [242] a wrist-worn solar-powered wearable platform has been designed and built. The platform can continuously monitor the photoplethysmogram (PPG) of the wearer during daily activities.

In addition to wrist watch, Dionisi *et al.* [243] proposed and designed a solar energy harvesting based self-powered autonomous T-shirt for measuring different vital signs, including respiration rate, heart rate, and body movement.

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A flexible solar panel with a dimension of 270mm × 175mm and 56g of weight is attached to the back of the T-shirt, to produce 19V and 100mA in open circuit and short circuit, respectively. Then, the harvested energy is used to power three different monitoring and communication modules. Their evaluation results have shown the feasibility of the prototype to be self-powered by solar energy harvesting. In [244], an autonomous body-worn wireless sensor node has been designed. The sensor node can harvest approximately 56 μW of electrical power using a flexible solar cell from indoor ambient light. Their experimental results shown that the prototype is able to continuously monitor the body temperature of the wearer, read, and transmit the data to a basement node for more than 15 hours without the need of battery.

Energy harvesting is gradually finding its way in wearable devices. It has shown great potential in mitigating the power constraints of today's battery-powered wearable devices, and in improving the operation life of many wearable applications. In this section we have surveyed the existing efforts in the design and development of energy harvesting-enabled wearable devices. Depending on the source where the ambient energy is harvested from, we focused on three different categories:**1) Kinetic Energy Harvesting**, **2) Thermoelectric Energy Harvesting**, and **3) Solar Energy Harvesting**, and discussed the most representative wearable prototypes in those categories. We can see that shoes are the most popular type of form in designing energy harvesting powered wearable devices [221]–[223], [225]. This is mainly because of the potential of high energy harvesting rate from the human heel strike. Another popular form of energy harvesting wearable devices is wrist watch, which is able to harvest energy from all kind of energy sources. Examples such as kinetic [218], thermoelectirc [230], [231], and solar powered watches [239]–[241]. Moreover, with the recent advancements in MEMs-based energy harvester, the form factor of energy harvester are shrunk in the order of millimetres. This makes it possible to embed those MEMS-based energy harvester into many tiny wearable devices that have very limited form factor.

The feasibility of utilising a particular energy harvesting technique for powering the wearable device is definitely application specific. It is hard to conclude which technique is more preferable than the others. For instance, for the kind of devices that will be carried or worn by the users during their daily activities, such as fitness trackers and smart-shoes, kinetic energy harvesting would be a good candidate, as more energy could be scavenged from the users' daily movement. However, for the application scenarios such as elderly health monitoring where the momentum of the user is low, kinetic energy is not a good choice. Instead, solar energy harvesting, could be very useful in those cases.

D. Summary of Energy Efficiency

Although battery technology has progressed over the years, it was only a linear progress compared to electronic devices which have shown an exponential growth [245]. Proposing new materials and architectures, in addition to improving surface structure, and chemical reactions, can advance the technology of batteries for wearable devices. On the other hand, minimising the power consumption of wearables through energy-efficient solutions is a very active area of research as it can extend the battery lifetime of wearables. However, users still need to recharge or replace the battery of their wearable at some point. For instance, the very recent Apple Watch can only supply power up to 18 hours of operation, and thus needs daily recharging [246]. The need for frequent recharging makes the wearable less desirable, especially in health applications such as aged-care, where the users may easily forget to recharge their wearables. Therefore, providing self-powered wearables through energy harvesting technology is highly desirable. Energy harvesting (EH) is a fast growing area which attracts considerable efforts from both research community and industry. We can expect that, future advancements in energy harvesting will change our way in powering wearable devices. Indeed, significant advancements have been recently made in the EH hardware technology leading to many off-the-shelf products available at low cost [247]–[249]. These developments point to future wearables and mobile devices that will be equipped with EH hardware to ease the dependence on batteries and potentially provide self-powered wearables [250].

V. Wearable Computing

In this section we discuss the wearable computing challenges. While energy constraints that were discussed in Section IV are essential in wearable computing, this section specifically focusses on research related to tackling the computation limitations in wearable devices. First, we discuss the *Offloading* techniques that either introduce dedicated hardware for certain tasks easing the load of the CPU or use cloud-based offloading in wearables. Next, under *In-device Machine Learning*, we discuss the recent approaches in implementing complex machine learning models such as deep learning models, directly in wearable devices by altering them to fit into limited resource footprints.

A. Offloading

Approaches aiming at offloading energy-hungry computational tasks to more energy efficient hardware or infrastructure have been widely proposed in the literature as a means of reducing energy consumption in wearables [251]. Although most of the work listed below are originally designed for smartphone platforms, such methods can also be adopted to wearable devices that are more constrained in both computation and energy sources.

1) Hardware Offloading: The main energy consumption in wearable computing is not only the cost of powering the sensors, but also the energy consumed by the microcontroller (MCU) for continuous data processing and computation. Thus, number of work looked into the possibility of introducing application specific low power processors to reduce the load on main CPU, which in return will also have savings in power consumption.

Priyantha *et al.* [252] proposed a heterogeneous multi-processor sensing architecture for smartphones. The

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proposed work exploits an additional low power co-processor to offload the sensing tasks from the energy hungry main processor, by doing so, the main processor can stay in low-power sleep mode for a longer period, while the co-processor can keep active and efficiently execute the repetitive sensing tasks. The evaluation results show that the proposed multi-processor architecture can achieve a two order-of-magnitude energy saving. Moving one step further, to achieve higher energy efficiency, in the work of Ra *et al.* [253], the authors investigated the partitioning of applications and sensing tasks across the main processor and low-power co-processor on mobile devices. The impacts and overheads of different offloading partitioning decisions have also been studied. Lu *et al.* [254] presented an energy efficient speaker identification system called *SpeakerSense*. The system enables energy efficient continuous audio sensing by leveraging the multi-processor architecture that splits the sensing and computation across the low-power co-processor, which continuously detects the surrounding sound, and power-hungry main-processor, which mainly used to process computational speaker identification task when triggered by the low-power co-processor. Another example which utilises multi-processor to achieve energy efficiency is the Dsp.ear [255]. Dsp.ear is an integrated sensing platform which can perform several context detection tasks, such as speaker counting, speaker identification, emotion recognition, and ambient sounds classification, using audio sensing pipelines. As shown in the results, by leveraging the support for DSP co-processor, the proposed work can achieve three to seven times increase in battery lifetime compared to conventional solutions using the main processor only. More recent work proposed *ZOE* [256], a wearable platform that supports three types of human-centric sensing including activity recognition, transportation mode detection, and user stress periods recognition. ZOE uses the Intel Edison SoC that includes a dual-core 500 MHz CPU, as the main processor, and a 100 MHz MCU, as the low-power co-processor. Moreover, in the design of ZOE, the Edison is coupled with a second MCU to form a 3-tier computational hierarchy, and thus, enable energy-efficient heterogeneous computation.

While hardware offloading is a preferred choice by both research community and industry as a means of offloading computational tasks, it alone can not address the complete spectrum challenges in wearable computing. First, the co-processor will not have the full computing power as the main CPU and as a result may not be suitable for all types of tasks. Second, co-processors come at the expense of valuable real state in device footprint and may not be suitable for all types of wearables, especially the ones that are limited in terms of size such as smart jewellery. Thus, cloud offloading is looked into as another means of increasing the effective computing power of the wearable devices as we next describe.

2) Cloud Offloading: As opposed to these hardware-based approaches, the software-based approach to achieve energy efficiency in wearable computing is to offload energy and time consuming computations to a remote server, such as the Cloud [257]–[259]. Many efforts have been made in exploiting cloud-based processing to reduce the energy consumption of mobile devices [260]–[264].

The *Cuckoo* framework [265] is designed to offload computation tasks on Android based handheld devices to cloud resources to reduce the energy consumption. The system provides a dynamic runtime mechanism to decide the execution location of the application. The *SociableSense* platform proposed by Rachuri *et al.* [261] investigates the use of machine learning algorithms to intelligently offload computational tasks, such as classification and inference algorithms, from local mobile devices to the cloud. Similarly, *CloneCloud* [262] utilises static analysis and dynamic profiling to automatically distribute application executions between local device and cloud server. By using an application-level virtual machine, the system can seamlessly offload part of the execution from the local mobile devices to the device clones running in the cloud sever, and thus, achieves up to 20 times speed-up in execution time and 20-fold energy saving on the mobile device. In [263], a novel GPS sensing approach *CO-GPS* was proposed. By offloading the GPS processing to the cloud, CO-GPS can leverage the existing information on the Web to achieve energy efficient GPS localisation, while the wearable devices only need to perform 2ms of GPS sensing. Their approach enables the GPS receiver to perform aggressive duty cycling and can reduce the energy consumption more than 3 orders of magnitude.

Apart from the above research work, cloud based approaches are extensively used in existing commercial wearables. For example, voice command service in available Android and Apple smart watches are cloud based. Similarly, number of real-time fitness feedback functions in wrist bands are also cloud based. The main disadvantage of this approach is the cost of communication. Increased communication, either directly with the cloud or via the smartphone will consume energy. Also, under bad network conditions latency will impact the user's quality of experience. Finally, that hardware required for communication will add to the device footprint and make it bulkier or users might need to carry the smartphone with the wearable to enjoy the full functionality. As a result, more recent work looked into the possibility of implementing some learning algorithms in the wearable itself limiting the communication with the cloud. We describe such work in the next subsection.

B. In-Device Machine Learning

Machine learning has become an integral part on wearables as it allows to make various kinds of inferences from the sensor data such as activity detection. Usually this is done by offloading the task to the cloud or to the paired smartphone. In some occasions this approach might not be useful. For example, consider a wristband which does not have any 3G/4G connection. In order to correctly detect the activities, the user always needs to carry the smartphone with her in addition to the wristband during workout sessions, which is inconvenient. Also, critical applications such as *fall detection* required fast decision making at the wearable rather than relying on the wireless channel to transfer data to the cloud. As a result, number of work looked into the possibility of implementing in-device machine learning models and altering existing

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machine learning models to fit into the resource constrained devices.

Computational complexity is a contributing factor to the power consumption of wearables because higher computational complexity leads to higher CPU usage and consequently higher power usage. For example, in human activity recognition application, when a large number of activities are to be classified, the model responsible for classification can be complex. Also, an imprudent selection of features can result in high power consumption. Therefore, reducing computational complexity in wearable computing can be achieved by reducing the feature set and the algorithm complexity.

Number of work investigated the possibility of reducing the feature set in a learning task so that the complexity of the problem reduces. Depending on the application, the approaches and the selected features are quite different. A comparison of various feature selection methods and classifiers was done by Khalifa *et al.* [266] to identify the minimal number of features to detect whether the user is using an escalator or a lift in multi-floor pedestrian navigation application. Zhang and Sawchuk [267] proposed a feature selection-based framework for human daily ambulation activities. They developed a multi-layer feature selection framework which takes advantage of the inherent structure of human activities and performs feature selection and classification in a hierarchical manner.

Saeedi *et al.* [268] proposed a cost-sensitive feature selection approach to minimise power consumption of on-body sensor localisation. They applied a greedy approach to find power optimised set of features by considering power consumption as a criterion while the performance is retained at a minimum level of accuracy. The proposed approach achieved an energy saving that ranges from 88% to 99.59% while obtaining an accuracy performance between 73.15% and 99.85%. Moreover, Palmerini *et al.* [269] developed a feature selection algorithm for posture analysis in Parkinson's Disease (PD). The developed algorithm was based on different classifiers and a nested cross validation, to maximise robustness of selection that better characterises the distinctive behaviour of PD and control subjects.

Genetic programming (GP) has been also used by many studies to find a set of discriminative and variation tolerant features that may reduce the energy requirements of the wearable device while maintaining the robustness of the targeted application. Genetic programming-based feature selection has been applied to many application areas such as ball bearing fault classification [270], activity recognition, and gesture recognition [271].

More recently, Ghasemzadeh *et al.* [272], introduced the notion of power-aware feature selection, which aims at minimising energy consumption of the data processing for classification applications such as action recognition. They proposed a graph model which takes into consideration the energy cost of individual features that are calculated in real-time and select the features in a power-efficient manner. Based on experimental results, they demonstrated that feature selection can significantly reduce energy consumption of the computing module, resulting in more than 30% energy savings while achieving high classification accuracy. On the other hand, Khalifa *et al.* [273] proposed an adaptive algorithm which is not only reducing the number of features to be used but also the computation complexity of the classifier. Their algorithm relies on the observation that the given set of activities to be recognised changes according to the current context of the user. Therefore, instead of using a single complex classifier based on a large set of features, their algorithm employs multiple simple classifiers each trained to classify only a subset of activities using a small number of features. Then at runtime, according to the current context, the system switches to the right simple classifier, minimising the computational complexity of classification.

In last few years *deep learning* methods evolved significantly as a branch of machine learning and found performing significantly better compared to conventional machine learning models in tasks such as image classification and speech recognition. Deep learning also eliminates the need to manually curating the features as its *representative learning* nature allows it to learn from raw data rather than from feature representations of data. Thus, wearables can be benefitted by using deep learning as a learning method for tasks such as activity recognition, voice commands, and object recognition. Nonetheless, training deep learning models is highly resource consuming and usually need number of GPUs and large memory footprints that may make them unsuitable for resource constrained devices. As a result, number of work evaluated the possibility of using deep learning models in wearables [274], [275] and proposed modified versions of deep learning models that are suitable for wearables [276].

Lane *et al.* [274] did a measurement study of two common deep learning models; Convolutional Neural Networks (CNNs) and Deep Neural Networks (DNNs), on representative mobile and embedded platforms. To represent wearables and other IoT devices, authors selected Intel Edison SoC. Authors measured the execution time, energy consumption, and memory footprint to make predictions using four pre-trained models. Results showed that Intel Edison SoC is able to make predictions using mid-size deep learning models such as AlexNet for image classification [277]. Nonetheless, such tasks were shown to push the processor to the limits. Bhattacharya and Lane [275] experimented with using another common deep learning model, Restricted Boltzmann Machines on a Snapdragon 400 SoC that is commonly used in wearables, and showed that even relatively simple RBM-based model outperforms conventional machine learning models such as SVM and Random Forests.

More recently, Lane and Bhattacharya [276] also proposed *SparseSep* that makes the weights of the fully connected layers of the deep neural network sparse and separation of convoluted layers to reduce the resource footprint. Authors evaluated the performance of *SparseSep* with four common processors used in wearables and showed that it allows efficient inferences from different deep models, on average requiring 11.3 times less memory and running 13.3 times faster on these representative platforms.

To summarise, while traditionally there are number of work in *in-device machine learning*, there is an increased interest in moving the state-of-the-art deep learning models (or parts of

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them) to wearables. This is an interesting research direction as deep learning methods are showing significant results in multiple domains and the possibility of porting them (fully or partially) to wearables will increase the possible features that can be offered in a wearable device. For example, such feature can be in-device fitness assistant and real-time speech translation. As the area is still evolving, we believe there is a significant research scope in bringing the state-of-the-art deep learning models to the wearables whilst carefully considering the trade-offs in accuracy and model complexity.

C. Summary of Wearable Computing

In summary, research directions that attempt to address the computing challenges faced by wearables can be fit into thee categories and each of them has its own advantages and disadvantages. *Hardware offloading* might increase the device footprint and may not be suitable for all the features offered by wearables. Nonetheless, hardware offloading can offer the lowest latency and high accuracy of the specialised tasks it is performing. According to the trends happening in smartphone domain it may be possible in future for the wearable devices to have machine learning/artificial intelligence specific hardware for functions such as face and speech recognition [278]. *Cloud offloading* can augment the CPU power of the wearable to any scale required by the application, yet comes at a cost of increased latency and communication costs.

The third approach of *In-device machine learning* sits in between in terms of having specialised hardware and utilising the full capability of the cloud by utilising the device's CPU and GPU, and by implementing more powerful machine learning models including state-of-the-art deep learning models yielding high accuracy. The approach in majority of such work is to take high performance machine learning models and import a compact of it (e.g., by making the model weights sparse) or parts of it (e.g., porting only the models weights of last few layers) so that an acceptable accuracy can be achieved according to the hardware footprint of the wearable. The approach can be further extended as highlighted in a recent research [279] where authors proposed a system called *Neurosurgeon* that dynamically partitions the neural network model between the cloud and the end device to achieve low latency and low energy consumption.

VI. Open Research Challenges and Future Directions

Wearable penetration is far from saturated. At the same time, there remains outstanding opportunities and challenges for the wearable technology. As uptake of wearables continue, we can expect evolution in products as well as new research directions. In this section, we discuss some potential new opportunities, open research challenges, and future directions for wearables. We believe that the wearable products and platforms themselves will continue to explore new shapes and formats trying to attract the market with even greater variety. On the research front, there will be new opportunities and challenges in medical use of wearables, communication security, energy efficiency, and utilisation of multiple wearables.

A. Trends in Wearable Products and Platforms

1) Wearable Products: As our product survey showed, current wearable market is dominated by *Accessories*, specifically *Smart watches* and *Wrist bands* in terms of device availability as well as sales. Market reports forecast that this trend is likely to be continued for next few years [9], [280]. In addition, *Smart Eye-wear* is expected to grow rapidly [281].

Nonetheless, our product survey showed that more recently a large number of *E-Textile* products are being released to the consumer market. Recent market reports corroborate the same trend [9], [280] and claims that *E-Textile* sales are expected to reach 15 million units by 2020. This is a 3800% increment in comparison to sales figures in 2015. To compare with, growth predicted for *Smart watches* is only 249% during the same time. Our survey on *research prototypes and systems* showed that there are only limited amount of research that leverages *E-Textile*. The current lack of research can be attributed to the limited availability of ready made consumer products in *E-Textiles*. However, with the predicted availability of *E-Textiles*, we believe number of research opportunities lies ahead to use them to build various systems, secondary application scenarios, or value-added services. On the one hand, compared to bands of various types and straps, *E-Textiles* can cover significant portion of the body area and is capable of taking measurements non-intrusively. On the other hand, *E-Textiles* has to be comfortable to wear, and must be conveniently washable, and must be visually pleasing and follow the fashion trends. Thus, a major challenge in future research related to *E-Textile* is to come up with a balance between this utility-usability trade-off. Similarly, *E-Patches* are also expected to show an exponential growth in the upcoming years, approximately 1900% growth from 2015 to 2020 [282].

2) Augmented Reality (AR): In addition to the current applications of wearables, augmented reality is becoming popular as an application, especially in *Smart Glasses*. Major applications include receiving information to a display (e.g., Google Glass, Vuzix-M) and interactions with surroundings (e.g., Icis, Waveoptics). However, the current commercial products are not yet mature enough to become largely popular and are worn only *"as and when required"* basis than normal daily use. Thus, the true potentials of wearable AR are yet to be explored.

As discussed in Section II-B, there is an increasing trend in the identification and modelling of physical objects in the surrounding environment through techniques such as gaze tracking [49], [155]. These developments enable the AR applications to interact with physical world and will fuel the development of realistic cyber-physical systems.

3) Third Party Apps and Services: The development and popularity of wearable devices are closely associated with the development of sensors and the services provided by apps. A large number of apps today are still first party apps, i.e., developed by the device manufacturers. First party apps often provide limited services. While some wearable ecosystems support third-party apps at the moment, the number of third-party apps are considerably lower compared to the smartphone ecosystems [283]. Hence, more third-party apps with higher degree of complexity can be expected in the future [284].

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TABLE III
CLASSIFICATION OF WEARABLES BASED ON THE APPLICATIONS

		Functions						
		Monitoring Vital Signals	**Communication & Notification**	**Ambient Sensing**	**Sensory / Haptic Applications**	**Payments**	**Gesture Detection**	**Virtual Reality /Augmented Reality**
Accessories	**Wrist-worns**							
	Smart watches	Apple Watch [30] Moto360 [32] Samsung Gear S2 [31] Pebble Time [33]	Apple Watch [30] Moto360 [32] Samsung G. S2 [31] Pebble Time [33]	Samsung G. S2 [31] Moto360 [32]		Apple Watch [30]	*Smart-watch Finger Writing [35]* Smart-watch Life Saver [34]	
	Wrist bands	FitBit Surge [37] Jawbone UP [36] LG VC110 [162] MOOV NOW [38]	Nymi Band [39]				*Wrist-worn Smoking Gesture Detector [41] Wrist-worn Bioimpedance Sensor [40] Ultrasonic-speaker Embedded Wrist Piece and Neck Piece [42]*	
	Head-mounted Devices							
	Smart eyewear	*Google Contacts [47]*	*Google Glass [46] Indoor Landmark Identification Supporting Wearables [50]*	FUN'IKI Glasses [44] *Google Glass [46]*			*iShadow Mobile Gaze Tracker [49]*	Hololens [43] HTC Vive [305] Recon Jet [45] FUNIKI Ambient Glasses [44] *Object Modelling Eyewear [48] Chroma [51]*
	Headsets & Ear-buds	LifeBeam Helmet [306]	Sony Xperia Ear [52] Samsung Gear IconX [99] Apple AirPods [53]		Proximity Hat [155]			
	Other Accessories							
	Smart jewellery	Bellabeat Leaf [57]	Smarty Ring [55]	CliMate [106]		Kerv [56] NFC Ring [105]	*Typingring [58] Gesture Detection Ring [59]*	
	Straps	Zephyr BioHarness [61]	MYO Amrband [60]		Active Protect Wear *Pneumatic Armband [62]*			
E-Textiles	**Smart garments**	Athos Shirt [64] Omsignal Shirt Citizen Sciences *MyoVibe [68] Dopplesleep [69]*		Viking Safe@Sea [126]	WarmX [129] Hug Shirt [65] T Jacket [128]	Contactless Jacket [130] Solar Shirt [66]		
	Foot / Hand-worns	Sensoria [71] Lechal Shoes [70]	Fujitsu Gloves [72] *Gait Analysis Foot Worns [74]*			GPS SmartSole [135]	*LookUp [73] Foot-worn Inertial Sensors [75]*	
E-Patches	**Sensor patches**	HealthPatch MD [76] Valedo [307]			Thync [77] UPRIGHT [78] Chrono Therapeutics [142] Smoking Cess. Patch [142]	Proteus [139]		
	E-Tattoo / E-Skin		Motorola E-tattoo [82] My UV Patch [143]			Electronic Fingertip [161] Wearable Stamp [83]		
	Others				Smart Tooth Patch [81]			

4) Direct Communication With the Internet: Majority of the existing wearables communicate with the Internet via a proxy device such as a smartphone or a personal computer. Earlier, this was due to the hardware resource constraints as many wearables were not equipped with long-range communication chipsets, e.g., WiFi or Cellular. Later, even though

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TABLE IV
ACCESSORIES

Accessories			
Sub Category	**Product**	**Description**	**Technical Specification**
Wrist-worns			
Smart watches	Apple Watch Series 2 (Aluminum case - 38mm) [30]	A smart watch with notifications and health tracking functionalities.	**CPU:** Dual-Core Processor. **GPU:** - **Internal Storage:** - **RAM:** - **Dimensions:** 38.6(h) x 33.3(w) x 11.4(d) (mm) **Weight:** 28.2 g **Display:** ION-X strengthened glass, second-generation OLED Retina display. Diagonal, 326 dpi (272 x 340). **Continuous Heart Rate Monitoring:** Yes **Battery Type/Capacity/Life:** Rechargeable Li-ion battery, Up to 18 hrs. **Charging Method:** Magnetic charging cable. **Sensors:** Accelerometer, gyroscope, ambient light sensor, heart rate sensor. **Operating System:** WatchOS 3. **Connectivity:** Bluetooth 4.0, Wi-Fi (802.11b/g/n 2.4GHz), NFC. **Compatibility:** iOS **Location Support:** GPS **Release Date:** 07/09/2016 **Price:** $349
	Motorola Moto 360 (42mm) [32]	A smart watch with notifications and health tracking functionalities.	**CPU:** Qualcomm Snapdragon 400 with 1.2 GHz quad-core CPU (APQ 8026) **GPU:** Adreno 305 with 450 MHz GPU. **Internal Storage:** 4 GB **RAM:** 512 MB **Dimensions:** 42 mm diameter by 11.4 mm high. **Weight:** 49 g **Display:** Corning Gorilla Glass 3 Backlit LCD, 1.37" (35 mm), 263 ppi (360 x 325). **Continuous Heart Rate Monitoring:** Yes **Battery Type/Capacity/Life:** Li-Poly/300 mAh/Up to 1.5 days of mixed use with Ambient off. Up to a full day of mixed use with Ambient on. **Charging Method:** Charging cradle. **Sensors:** Accelerometer, ambient light sensor, gyroscope, vibration/haptics engine. **Operating System:** Android wear. **Connectivity:** Bluetooth 4.0, Wi-Fi (802.11b/g/n 2.4GHz), NFC. **Compatibility:** Android/iOS **Location Support:** No **Release Date:** 14/09/2015 **Price:** Starting from $349.99
	Samsung Gear S2 [31]	A smart watch with notifications and health tracking functionalities.	**CPU:** Exynos 3250, Dual-core 1GHz Pega-W **GPU:** - **Internal Storage:** 4 GB **RAM:** 512 MB **Dimensions:** 42.3 x 49.8 x 11.4 (mm) **Weight:** 47 g **Display:** Full Circular SAMOLED, 302 ppi (360 x 360). **Continuous Heart Rate Monitoring:** Yes **Battery Type/Capacity/Life:** Li-ion/250 mAh/Up to 3 days **Charging Method:** Wireless charging dock **Sensors:** Accelerometer, gyroscope, hear rate Monitor, barometer, ambient light sensor **Operating System:** Tizen Based Wearable OS 2.3.1. **Connectivity:** BT4.1, Wi-Fi, NFC **Compatibility:** Devices running Android 4.4+, iOS 9.0+. **Location Support:** GPS **Release Date:** 2015 **Price:** $250-350
	Huawei Watch (42mm) [86]	A smart watch with notifications and health tracking functionalities.	**CPU:** Qualcomm Snapdragon 400, APQ8026 Quad 1.2GHz **GPU:** - **Internal Storage:** 4 GB **RAM:** 512 MB **Dimensions:** 42 x 11.3 (mm) **Weight:** 96-134 g (with strap) **Display:** 1.4-inch full circle AMOLED display, 286 ppi (400x400), 10,000:1 high contrast ratio **Continuous Heart Rate Monitoring:** Yes **Battery Type/Capacity/Life:** Li-ion/300 mAh/1.5-2 days **Charging Method:** Charging cradle **Sensors:** 6-axis motion sensor (gyroscope + accelerometer), heart rate sensor (PPG), barometer, vibration Motor. **Operating System:** Android Wear **Connectivity:** Bluetooth 4.0, Wi-Fi (802.11b/g/n 2.4GHz), NFC. **Compatibility:** Devices running Android 4.3+ or iOS 8.2+ **Location Support:** No **Release Date:** 02/09/2015 **Price:** $149-349

more and more wearable devices supporting WiFi connectivity, there is significant gap in the adaptation of direct Internet communication by wearables. For instance, majority of fitness tracking wearable applications still rely on the smartphone for the communication with cloud servers. The potential reasons may be the lack of support in wearable operating systems to enable secure direct communication and/or the slower growth in third-party app development for wearables [196]. However, cellular and/or WiFi enabled wearables are continuously increasing [284], [285] and there is an increasing demand for devices that can directly connect with the Internet [286].

5) Web of Wearables: Some of the surveyed wearables such as *Moto360* and *Google Glass* provide Web access to the users. Nonetheless, Web pages are not currently in a optimal design to be displayed in wearables and the different shapes

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TABLE V
ACCESSORIES (CONTINUED)

Accessories			
Sub Category	**Product**	**Description**	**Technical Specification**
Wrist-worns			
Smart watches	FitBit Surge [37]	A fitness tracker.	**CPU:** 32-Bit ARM Cortex-M3, 48MHz. **GPU:** - **Internal Storage:** 1024KB Flash. Saves 7 days of detailed motion data minute by minute. **RAM:** 128 KB **Dimensions:** 20.88 x 24.36 x 34 (mm) **Weight:** 2.7 ounces **Display:** Touch screen, monochrome LCD, LED backlight for low light visibility. 1.26" Diagonal, 175 ppi (144 x 168). **Continuous Heart Rate Monitoring:** Yes **Battery Type/Capacity/Life:** Li-Poly/-/Up to 5 days (up to 10 hrs with GPS). **Sensors:** GPS, 3-axis accelerometers, 3-axis gyroscope, digital compass, optical heart rate monitor, altimeter, ambient light sensor, vibration motor **Operating System:** **Connectivity:** Bluetooth 4.0, Wi-Fi (802.11b/g/n 2.4GHz), NFC. **Compatibility:** Syncs with leading smartphones, tablets and computers running iOS, Android or Windows. **Location Support:** **Release Date: Price:** $249.95
	Pebble Time [33]	A smart watch with notifications and health tracking functionalities.	**CPU:** ARM Cortex M7 processor **GPU:** - **Internal Storage:** 16 MB **RAM:** **Dimensions:** 40.5 x 37.5 x 10.5 (mm) **Weight:** 64.6 g **Display:** 200 x 228 pixels **Continuous Heart Rate Monitoring:** Yes **Battery Type/Capacity/Life:** -/-/Up to 10 days **Charging Method:** Charging dock **Sensors:** Microphone, heart rate monitor, accelerometer, pedometer **Operating System:** Pebble OS 4.0+. **Connectivity:** Bluetooth 4.0, Wi-Fi (802.11b/g/n 2.4GHz), NFC. **Compatibility:** Devices running Android 4.3+ or iOS 8.0+ **Location Support:** Not mentioned. **Release Date:** 30/06/2016 **Price:** $169
	Smart-watch Life Saver [34]	A smart watch application that assists untrained people with performing Cardiopulmonary Resuscitation (CPR).	
	Finger-writing with Smart-watch [35]	The system accurately identified finger, hand and arm gestures by monitoring movements of the tendons passing through the wrist using the accelerometer and gyroscope sensor attached to a smart watch.	

and sizes of various wearable screens make this more complicated. Similar to the *mobile Web* standards/guidelines and validating scheme to assess the readiness of content for the mobile Web proposed by W3C [287], there is a requirement for similar measures for wearables. The major challenge is how to extract only the most relevant information from a Web page and show in a user friendlier manner in wearables such that interactions and navigation is effortless.

B. Medical Use of Wearables

Health has been one of the major motivations for the adoption of wearables. Wearables of today, however, are mainly used for fitness and well-being application by the users. Physicians still do not use or "trust" the data produced by consumer wearables. First, wearables use consumer hardware to sense pulse rate, heart rate and other health related data. These data therefore have "consumer-grade accuracy", which makes it difficult for physicians to use them for critical diagnoses. How to improve the accuracy of sensing health-related data without blowing up the hardware budget, therefore would be a critical new challenge to solve. Second, there are many other health data including photographic details of internal organs, that can only be collected with more invasive medical sensors or implantable medical devices (IMDs). Therefore, the seamless communication with such IMDs to receive, store and

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TABLE VI
ACCESSORIES (CONTINUED)

Accessories			
Sub Category	**Product**	**Description**	**Technical Specification**
Wrist-worns			
Wrist bands	Sony Smartband 2 [308]	An activity tracker, sleep tracker and smart coach that also enables cardless payments.	**CPU:** ARM Cortex-M0 32-bit processor **GPU:** - **Internal Storage:** 256KB internal embedded flash memory **RAM:** 32KB **Dimensions: Weight:** **Display:** 3 LED indicators. **Continuous Heart Rate Monitoring:** Yes **Battery Type/Capacity/Life:** Li-Poly/100 mAh/Up to 2 days under normal mode. **Charging Method:** Micro USB charging. **Sensors:** Accelerometer, heart rate monitor. **Connectivity:** Bluetooth 4.0 (transfer rate < 3Mbps), NFC. **Compatibility:** Android devices running Android 4.4+, iPhone running iOS 8.2+. **Release Date:** 29/09//2015 **Price:** $80
	Mi Band 2 [309]	An activity tracker that can also be used to receive notifications and unlock a phone.	**Dimensions:** 15.7x 10.5 x 40.3 (mm) **Weight:** 19 g **Display:** OLED display **Battery Type/Capacity/Life:** Li-Poly/70 mAh/20 days. **Charging Method:** USB charger. **Sensors:** Accelerometer, optical heart rate sensor. **Connectivity:** Bluetooth 4.0 BLE. **Compatibility:** Devices running Android 5.0 and later. **Location Support:** No **Release Date:** 07/06/2016 **Price:** $23-43, depending on the seller.
	LG Gizmopal 2 [162]	A wrist band equipped with GPS location sensing and cellular capabilities, designed for enabling calls between a wearer and caregivers.	**CPU:** QSC6155 **Internal Storage: RAM:** **Dimensions:** 2.17"(H) x 1.37"(W) x 0.55"(D) **Weight:** 1.60 oz (without wrist band) **Display:** LED **Battery Type/Capacity/Life:** -/510 mAh/2.5 hrs usage time, 9 days standby time. **Charging Method:** Micro USB charging. **Sensors:** Location tracking sensors through Stand-Alone, AGPS, and WPS. **Connectivity:** CDMA, 1.9 GHz CDMA PCS, 800 MHz CDMA. **Compatibility:** Devices running Android 4.0+ and iOS 7.0+. **Location Support:** Location tracking sensors through Stand-Alone, AGPS, and WPS. **Release Date:** 06/11/2015 **Price:** $79.99
	FitBit Flex 2 [310]	A fitness tracker.	**Internal Storage:** Saves 7 days of detailed motion data minute by minute, saves daily totals for past 30 days. **RAM:** **Dimensions:** 0.44" as the width of the sensor core. **Weight:** 23.5 g **Display:** Touch screen, monochrome LCD. **Battery Type/Capacity/Life:** Li-Poly/-/Up to 5 days. **Charging Method:** Charging cable with pins. **Sensors:** 3-axis accelerometer, vibration motor. **Operating System:** **Connectivity:** Bluetooth LE **Compatibility:** Devices running Mac OS X 10.6 and up, iPhone 4S and later, iPad 3rd generation and later, Android 4.3 and later, and Windows 10. **Location Support:** **Release Date:** 29/08/2016 **Price:** $99.95

display accurate medical data to the user or to the physician would be a potential future direction for wearables.

Existing wearable products are designed to communicate with mobile devices, such as smartphones or point-of-sale machines, or infrastructure equipment, such as wireless access points. As such, current versions are only equipped with popular mobile communication interfaces, such as NFC, Bluetooth, and WiFi. IMDs, however, employ different radio frequencies compared to the common mobile communication frequencies, primarily due to different propagation properties exhibited within human body as well as for health and safety issues. For example, Federal Communication Commission (FCC) rulings confirm that 402-405MHz frequency range is the most suitable for communicating through human body [288]. Consequently, current wireless endoscopy capsules [289], which take photos inside the bowel and transfer wirelessly to an outside recorder, use different frequencies compared to NFC, Bluetooth, and WiFi. Thus, the communication between future IMDs and consumer wearables remains an interesting open challenge.

C. Communication Security

As we highlighted in Section III-A, a range of communication-related security vulnerabilities that affect the operations and performance of wearables has been identified. Majority of the vulnerabilities are implementation issues than the limitations of technologies, where certain protocol requirements are not implemented due to performance concerns.

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TABLE VII
ACCESSORIES (CONTINUED)

Accessories			
Sub Category	**Product**	**Description**	**Technical Specification**
Wrist-worns			
Wrist bands	Empatica E4 Wristband [93]	It monitors physiological stress, arousal, sleep and physical activity.	**CPU: - GPU: -** **Internal Storage:** 60+ hrs flash memory. **RAM:** **Dimensions:** $44(l) * 40(w) * 16(h)$ (mm) **Weight:** 25 g (form factor) **Display:** No display except for an LED indicator. **Battery Type/Capacity/Life:** Li-Poly/260 mAh/36+ hrs under recording mode, 20+ hrs under streaming mode. **Charging Method:** Charging cradle. **Sensors:** PPG sensor, 3-axis accelerometer, EDA sensor (electrdermal activity sensor), infrared thermopile **Operating System: -** **Connectivity:** Bluetooth 4.0 **Compatibility:** Devices running iOS 8.2+ and Android 5.0+. **Release Date:** 04/03/2015 **Price:** $1690
	NYMI Data [39]	It creates a Personal Authentication Network.	**CPU:** CORTEX M4 PROCESSOR **GPU:** **Internal Storage: - RAM:** 2 MB **Dimensions:** $18.4(w)*2(t)$ (mm) with 4 sizes of length(S/M/L/XL) **Weight:** **Battery Type/Capacity/Life: -/-/**5 days. **Charging Method:** Charging cradle. **Sensors:** Accelerometer, ECG sensor system, vibration motor. **Operating System:** **Connectivity:** Bluetooth 4.1 BLE, NFC **Compatibility:** Devices running Android Marshmallow, Android Lollipop, iOS 9, iOS 10. **Release Date:** 09/2016 **Price:** $199
	Wrist-worn Bioimpedance Sensor [40]	An unobtrusive user identification system which identifies a certain user and detects if the device is on the body by measuring bioimpedance.	
	Wrist-worn Smoking Gesture Detector [41]	A machine learning algorithm to detect smoking gestures using wrist-worn 9-axis inertial measurement unit.	
	Ultrasonic-speaker Embedded Wrist Piece and Neck Piece [42]	An activity and context recognition method that requires the user to wear ultrasonic speakers on the wrist and a receiver on the chest/neck.	
	Pneumatic Armband [62]	It is an armband that generates sensations such as tapping, holding and tracing though air inflation and deflation.	

Thus, there is room for new research directions along the lines of clean-slate designs in some of the aspects in secure communications. We below summarise some research directions that show promise in wearable communications security and device security.

1) Service Discovery and Key Distribution: The device discovery and service discovery procedures between wearables and its controllers, i.e., a smartphone, require a device identity and/or a service identity to be advertised. However, this makes it possible to track users by eavesdropping the wireless channel. A possible approach is to design a secure discovery mechanism so that the wearable's identification is hidden from unintended targets.

Majority of wearables establish secure communications via pairing with smartphones or other controller devices. The pairing methods ensure secured future connections between wearables and their controllers. However, the information exchanging during pairing procedure, i.e., device I/O capabilities, secret key etc., may be collected by eavesdroppers or sniffing tools. In addition, a MITM attacker is able to force peer devices to use the weakest pairing method by actively modifying the information exchanged between peer devices. Therefore, another important issue to consider is the growing complexity of MITM attacks and sniffing tools. It requires robust security solutions to be proposed to prevent this issue, which can be a novel pairing model or added features to current pairing methods.

2) Wearable System Security: Security for data storage and data access is also an open challenge for wearable systems whether it is the local storage, controller or at the cloud server level. At any level, the data should be stored as encrypted information. A user or an app that wishes to access the stored

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TABLE VIII
ACCESSORIES (CONTINUED)

Accessories Cond.			
Sub Category	**Product**	**Description**	**Technical Specification**
Head-mounted Devices			
Smart eyewear	Microsoft HoloLens [43]	Performs holographic computing and generates displays of mixed reality.	**CPU:** Intel Cherry Trail SoC **GPU:** Microsoft Holographic Processing Unit (HPU) **Internal Storage:** 64GB eMMC **RAM:** 8MB SRAM, 1GB LPDDR3. **Operating System:** Windows 10. **Battery Type/Capacity/Life:** Rechargeable/-/-. 2-3 hours of active use, or 2 weeks of standby time. **Weight:** 579g **Connectivity:** WiFi 802.11ac, micro-USB 2.0. **Sensor:** IMU environment understanding sensors, one depth camera ($120^\circ * 120^\circ$ angle of view), one 2.4 MP video camera, one four microphone array, one ambient light sensor. **Release Date:** 21/01/2016 **Price:** $3000
	FUNIKI Ambient Glasses [44]	Displays smartphone notifications using LEDs in four available colors. The light also enriches the wearer's facial expressions to show your feelings or the atmosphere of the surrounding area.	**Battery Type/Capability/Life:** Li-ion battery/-/- **Charging Method:** USB charging. **Sensors (Actuators):** 1 Mini speaker, an acceleration sensor, 6 Full-color LEDs. **Weight:** 32g. **Connectivity:** Bluetooth 4.0 Low Energy. **Release Date:** 15/01/2016 **Price:** $242
	Recon Jet [45]	A pair of glasses equipped with various sensors, multiple connectivities which enables a wide range of functionalities.	**CPU:** 1 GHz dual-core ARM Cortex-A9. **Internal Storage:** 8 GB flash. **RAM:** 1 GB SDRAM. **Display:** Widescreen 16:9 WQVGA display. **Sensors:** 3D accelerometer, 3D gyroscope, 3D magnetometer, pressure sensor, infrared (IR) sensor. **Battery Type/Capability/Life:** Swappable Li-ion/-/Up to 4 hours of typical use per battery. **Charging Method:** USB charging. **Connectivity:** GPS, Bluetooth 4.0 (Bluetooth Smart), ANT+, Wi-Fi (IEEE802.11a/b/g/n), Micro USB 2.0. **Release Date:** 2015 **Price:** $499
	Google glass [46]	A wearable computer designed as glasses. It allows user interaction through a touchscreen and voice control and it is able to take photos and videos.	**Battery Type/Capacity/Life:** Li-ion/570 mAh/-. **Weight:** 36g. **Connectivity:** Wi-Fi 802.11b/g, Bluetooth, micro USB. **Sensor:** Bone conduction transducer, accelerometer, gyroscope, magnetometer, ambient light sensor, proximity sensor, touchpad, camera.
	Google Contact Lens [47]	The contact lens can detect the wearer′s internal body temperature, glucose and blood-alcohol content.	
	Object Modelling Eye-Wear [48]	A hand-free and feedback-free way of creating 3D models by analysing the gaze data of a user.	
	iShadow Mobile Gaze Tracker [49]	iShallow is a real-time eye-tracking platform with low power consumption. It saves energy by sub-sampling the pixels.	
	Indoor Landmark Identification Supporting Wearable [50]	Assiting landmark identification of blind people by eleiminating *masking*, which is a harmful interference in existing ETA and EOA (Electronic Orientation Aids).	
	Chroma [51]	Chroma is an augmented-reality application running on a Google Glass platform to support color-blind people.	

data at any level is required to be authenticated. Therefore, new lightweight user authentication mechanisms appropriate to wearables, which are usually limited in computational capability, may also be designed.

Wearables are mainly used to monitor user activities, health status and more sensitive personal information. A wearable device also needs to be authenticated by owner before he/she carries it and allow its functions to be activated. Thus new, user-friendly, and efficient device authentication mechanisms are also required to be built into wearables. In addition, a trust verification method specified for wearables would be a good solution for verify the reliability of a wearable device and the services provided by wearable systems.

Encryption methods are the most effective means to protect transmitted data within a wearable system between wearables and smartphone and from smartphone to cloud

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TABLE IX
ACCESSORIES (CONTINUED)

Accessories Cond.			
Sub Category	**Product**	**Description**	**Technical Specification**
Head-mounted Devices			
Head-worns	Sony Xperia Ear [52]	It uses Sony′s voice technology to take voice commands, taking calls, dictating messages and searching the internet.	**Dimensions:** $29.5 * 15.2 * 25.2$ (mm) (core unit), $29 * 41 * 60$ (mm) (charging case). **Weight:** 6g (core unit), 39g (charging case). **Battery Type/Capacity/Life:** Li-ion/65 mAh (core unit), 300mAh (charging case)/Up to 4.0 hours (talktime), up to 16 hours (talktime without charging case), up to 80 hours (standby). **Charging Method:** Charging case. **Sensors:** Proximity, Accelerometer, Gyroscope. **Connectivity:** NFC and Bluetooth 4.1. **Compatibility:** Android version 4.4 KitKat or later. **Release Date:** 13/12/2016 **price:** $199.99
	Apple AirPods [53]	A pair of wireless headsets that can pair with most other Apple devices, take voice commands via Siri, and detect if they are in your ears.	**Chip:** Apple W1 chip **Dimensions:** $16.5*18.0*40.5$ (mm) (each AirPod), $44.3*21.3*53.5$ (mm) (charging case) **Weight:** 4g (AirPods), 38g (charging case) **Battery Type/Capacity/Life:** -/-/More than 24 hrs (with charging case), up to 5 hrs of listening (one charge). **Sensors (of each AirPod):** Dual optical sensors, motion accelerometers, voice accelerometer, dual beamforming microphones, infrared sensors. **Connectivity:** Bluetooth (AirPods), Lightning connector (charging case). **Compatibility:** iPhone/iPad/iPod (iOS 10 or later), Apple Watch (watchOS 3 or later), and Mac (macOS Sierra or later), with instant switching between the devices. **Release Date:** 13/12/2016 **Price:** $159
	Samsung Gear IconX [99]	A pair of wireless headsets that tracks a wearer's speed, distance, heart rate and calories. It provides a voice guide and saves up to 1,000 songs.	**Internal Storage:** 3.5 GB **Dimensions:** $18.9 * 26.4 * 26.0$ (mm) (Earbud), $30.3 * 92.0 * 35.3$ (Case). **Weight:** 6.3 g (Earbud), 52 g (Case). **Battery Type/Capacity/Life:** -/47 mAh/Up to 3.8 hours standalone music play (Earbud), -/315 mAh/-(Case) **Charging Method:** Charging case. **Sensors:** Accelerometer, HR Sensor. **Connectivity:** USB 2.0, Gear IconX PC Manager, Bluetooth v4.1 (A2DP, AVRCP, HFP). **Compatibility:** Android 4.4+ **Price:** $199
	Jabra Elite Sport [100]	A pair of wireless headsets that enables a wide range of voice controls and fitness trackings.	**Dimensions:** Multiple sizes available. **Battery Type/Capacity/Life:** -/-/Up to 3 hours continuous play time. **Charging Method:** Charging case. **Sensors:** Clinical grade in-ear heart rate monitor, tri-axis accelerometer. **Connectivity:** Bluetooth v4.1. **Compatibility:** iOS, Android, Windows. **Price:** $249.99
	Bragi Dash Pro [54]	A pair of wireless headsets that enables a wide range of voice controls, high quality audio play and fitness tracking.	**Internal Storage:** 4 GB **Dimensions:** Multiple sizes available. **Battery Type/Capacity/Life:** -/-/Up to 5 hours playtime. **Charging Method:** Charging case. **Connectivity:** Bluetooth **Compatibility:** Any Bluetooth device. **Price:** $329
	LifeBEAM Smart Helmet [306]	It senses heart rate and calories consumption	**Battery Type/Capacity/Life:** Li-Poly/170mAh/Up to 15 hrs. **Weight:** 280g. **Connectivity:** Bluetooth LE and ANT+. **Sensor:** Optical heart-rate sensor & 3-axis accelerometer.
	Muse headband [311]	The headband measures brainwave signals with 7 calibrated sensors and provides real time feedback to the wearer during meditation process.	**Battery Type/Capacity/Life:** Li-Ion/-/-. Maximum 10hrs of use. **Connectivity:** BLE. **Sensor:** Brainwave signal sensors. **Weight:** 60g. **Price:** $250
	Joule [312]	A smart earring that tracks heart rate, calories, and activity level by leveraging the PPG (photoplethysmogram) technology.	**Connectivity:** Bluetooth 4.0. **Price:** $88

servers. However, it is worth noting that not all communication standards provide encryption for all data transmissions, which raises significant vulnerabilities in data confidentiality and integrity. Thus, future research work may pay more attention on design lightweight and low overhead encryption methods for communication of wearables.

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TABLE X
ACCESSORIES (CONTINUED)

Accessories			
Sub Category	**Product**	**Description**	**Technical Specification**
Others			
Smart jewellery	NFC Ring [105]	To unlock doors and smartphones, transfer information, and link people. It never needs charging.	**Dimensions:** $d >= 15.27$ (mm). **Charging Method:** Not required. **Connectivity:** NFC. **Compatibility:** Android devices. **Price:** $19.5-45.5
	Smarty Ring [55]	It′s tiny LED display shows notifications from smartphone, triggers the camera and allows users to make outgoing calls, controls music playing.	**Charging Method:** wirelessly. **Connectivity:** Bluetooth 4.0.
	Nod [313]	The Nod Backspin can be used to control drones or interact with objects during a virtual reality experience using hand wave gestures. It also features a game controller with a joystick and a set of buttons.	**Charging Method:** USB charging. **Battery Type/Capacity/Life:** -/-/Approximately 4 hours of active usage. **Connectivity:** Bluetooth.
	Motiv Ring [314]	A smart ring that tracks activities, heart rate, and sleep.	**Dimensions:** $8 * 2.5$ (mm) **Charging Method:** USB Magnetic Keychain Charging. **Battery Type/Capacity/Life:** Li-ion/-/3-5 days. **Sensors:** Optical heart rate sensor, 3-axis accelerometer, smart LED sensor. **Connectivity:** Bluetooth. **Compatibility:** iOS 9.0 or later. **Release Date: Price:** $199
	Bellabeat Leaf [57]	A clip / necklace / bracelet that tracks the movement of the wearer. Bellabeat App gives information on daily activity, sleep, and periods.	**Battery Type/Capacity/Life:** Replaceable coin cell battery CR2032/-/Last for 6 months. **Connectivity:** Bluetooth. **Price:** $119-199
	CliMate Clip-on [106]	It tracks the ambient humidity, UV rays and temperature and sends data to an iOS/Android platform regularly and report to alert users of sudden changes in weather.	**Battery Type/Capacity/Life:** Coin cell battery/-/Lasts up to 6 months.
	Mersiv [315]	A neckpiece-like device that assists the wearer with language learning by learning from the surrounding area as infants. The device captures sound and sights and pass the data to a phone for analysis and creation of personalized language lessons. The device can also communicate with the user through the built-in microphone under the interactive mode.	**Charging Method:** A charging base. **Connectivity:** BLE. **Sensors:** Camera, microphone, headphones.
	Kerv [56]	Can be used to make payments, validate on public transport, and unlock smart locks.	**Charging Method:** No charging.
	Typingring [58]	A wearable ring that enables text inputs without having a keyboard.	**Charging Method:** No charging. **Connectivity:** No pairing. **Price:** $130
	Gesture Detection Ring [59]	A remote gesture input solution using a wearable ring. The wearable ring also opportunistically harvest energy from an NFC-enabled phone.	

The wearables may provide real time or continuous data streams for specific services. As discussed earlier, there is a high potential that more wearables will be worn by the same person. Reducing radio channel interference of among wearables will also be required to guarantee the accuracy and real time transmission of collected data. Appropriate scheduling mechanism in communication standards may be designed to minimise interference or data collision at the data aggregating point, i.e., smartphone and/or cloud server.

3) Communication Protocols: Our survey showed that almost all the consumer wearables support only short range communication protocols such as Bluetooth and WiFi. Only a limited number of device supported direct 3G and 4G connections. With the advancements in IOT, a different set of protocols offering low power and long range communication are emerging. Such protocols include LoRA, SigFoX, and 802.11ah. These protocols might be adapted by some wearables and will open up some research directions in evaluating their performance with respect to security and privacy.

D. Energy Efficiency

It is expected that future wearables will rely on energy harvesting to be self-powered. Despite some work is available on the subject, still there is a gap between the amount of energy that can be generated and the amount required to power the wearables. This is particularly true with the function-rich wearables with various embedded power hungry sensors. Several recent attempts made inroads to further reduce this gap, moving us closer towards self-powered autonomous wearables.

1) Using Multiple Energy Harvesting Sources: Harvesting energy from multiple sources of environmental energy, such

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TABLE XI
ACCESSORIES (CONTINUED)

Accessories			
Sub Category	**Product**	**Description**	**Technical Specification**
Others			
Straps	Zephyr BioHarness [61]	Heart rate and breathing rate sensing.	**Dimensions:** $28(d) * 7(t)$ (mm). **Sensor:** 3-axis accelerometry, EC38 Type 3 ECG. **Release Date:** 27/07/2011 **Price:** $529
	Quell Relief [316]	This drug free OptiTherap technology automatically adjusts stimulation intensity to ensure the patient receives optimal pain relief.	**Battery Type/Capacity/Life:** Li-ion/-/Up to 30 hours per charge. **Dimensions:** $98 * 74 * 11$ (mm). **Weight:** 62g. **Connectivity:** Bluetooth. **Price:** $249
	Wahoo Tickr X [317]	This is a chest strap for heart rate monitoring.	**Connectivity:** Bluetooth 4.0 and ANT+. **Dimensions:** $2.75 * 0.5 * 1.5$ (inches). **Weight:** 8.5g. **Price:** $79.95
	Qardio Core [318]	It tracks EKG/ECG, body temperature, respiration rate, heart rate and assesses stress level.	**Battery Type/Capacity/Life:** Li-ion polymer/-/- **Dimensions:** $185 * 87 * 9$ (mm). **Weight:** 130g. **Price:** $800
	AMPY Move [319]	It generates energy by harvesting kinetic energy from a wearer's motion to power other IoT devices.	**Battery Type/Capacity/Life:** -/1800 mAh/- **Dimensions:** About the size of a deck of cards. **Weight:** About the weight of a smartphone. **Transformers:** Inductors. **Price:** $75
	MYO armband [60]	Myo reads gestures and motion to enable seamlessly control of presentation software.	**Charging Method:** Micro-USB charging, built-in rechargeable lithium ion battery, one full day use out of single charge. **Dimensions:** Expandable between $7.5 - 13$ inches. **Connectivity:** Bluetooth 4.0. **Price:** $199

as mechanical, thermal, and solar energy, simultaneously is a recent trend to improve the power generation performance to enable self-powered devices (hybridisation of energy harvesters) [290]. Some hybrid cells have been proposed for harvesting two or more sources such as mechanical and solar energy [291]–[295]. For integrated energy harvesters of two or more such sources, integrated electronic designs should be developed to accommodate different materials, different operation frequencies, and different generated outputs to achieve the maximum power output. This can help reducing the gap between the generated energy and the required energy enabling self-powered wearables.

The predicted use of multiple wearables also provides multiple opportunities to harvest energy from multiple body parts. The amount of energy that can be harvested from different body parts depends a lot on the user activity. For example, *Smart Shoe* can be the best wearable to harvest mechanical energy when a user is running, but not the best wearable when the user is cooking. These harvesting behaviours would trigger energy sharing mechanisms among the wearables either in the form of wireless energy transmission or distributed energy utilisation.

2) Using Energy Harvester As Sensing Units: In addition to exploiting different energy harvesters as energy supply to power embedded sensors, researchers have recently investigated the use of energy harvester as the sensing unit to eliminate the necessity of many energy hungry sensors. This can be considered an important future direction in reducing the gap between generated energy and required energy. Randall *et al.* [296] have investigated the idea of using wearable solar cells for indoor positioning. In this work, the solar cells are attached to the user's shoulder to harvest energy from indoor ambient light, in addition, the irradiance of the solar cells are used as the context information to support indoor position tracking. Their experiment results shown that a distance estimation accuracy of 21 cm can be achieved.

Khalifa *et al.* [297] proposed to use the harvested power signal from the kinetic energy harvester as the signal source for activity recognition. Instead of using accelerometer, they have demonstrated that the features extracted from the power signal of the kinetic energy harvester contains sufficient information to recognise common activities. Their experiment result has shown a 83% of accuracy in average, over 14 different activity sets. Other examples of utilising the voltage signal from kinetic energy harvester for context detection, such as step detection [298], calorie expenditure estimation [299], hotword detection [300], transportation mode detection [301], user authentication [302]. These recent research studies promise the realisation of self-powered wearables which continue to sense and monitor the user on a permanent basis.

E. Utilisation of Multiple Wearables

According to predictions, it is highly likely that individuals wear more than one wearable during their day-to-day life [303]. This will provide the ability to develop innovative services utilising the storage, processing, connectivity and sensing capabilities available through multiple wearables. Even today, wearables are well equipped for collaborating among each other and forming a personal wearable network.

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TABLE XII
E-TEXTILE

E-Textile			
Sub Category	**Product**	**Description**	**Technical Specification**
Smart garments	Hug Shirt [65]	The actuators recreate the sensation of touch, warmth and emotion of the sender to the Hug Shirt worn by the receiver.	**Charging Method:** Magnetic charger. **Connectivity:** Bluetooth. **Sensors:** sensors that feel the strength, duration, and location of the touch, the skin warmth and the heartbeat rate of the sender.
	T Jacket [128]	Designed for children with ADHD or Autism. It simulates deep touch pressure as hugs with controllable levels.	**Charging Method:** USB charging. **Actuator:** Air pressure actuators. **Connectivity:** Bluetooth. **Compatibility:** iOS 7.0+, Android 4.3+. **Price:** $399-599
	Solar Shirt [66]	The Solar Shirt uses solar cells integrated into the shirt′s fabric. The energy can be used to charge a USB compatible device, such as a smartphone, MP3 player or camera.	**Sensors:** solar panels and flexible electronics.
	Fabric Touch Panel [320]	Made of ultra-flexible organic field-effect transistors.	**Sensors:** Variable capacitors measuring voltage difference.
	Hovding [321]	A helmet that senses the cyclist′s movement patterns and reacts in case of an accident.	**Sensors:** Gyroscopes and accelerometers. **Battery Type/Capacity/Life:** -/-/Up to 18 hrs after fully charged. **Charging Method:** Micro USB charging. **Release Date:** 03/2015 **Price:** $335
	Viking-Ohmatex Smart Fire Fighter Gear [126]	The LED displays on the outfit gives the advantage to visually indicate critical heat levels and potentially critical situations.	**Sensors:** Thermal sensors embedded directly to the inner and outer fabric layers.
	Elastic Electronic Textile [322]	It is a research product designed by researchers in University of Tokyo. Circuits printed in conductive ink for EMG sensing.	**Sensors:** EMG sensor.
	Athos [64]	These products mainly target athletes and sports personnels as training aids.	**Sensors:** EMG, heart rate, breathing rate sensors. **Connectivity:** Bluetooth. **Price:** $348
	OmSignal [113]	Fitness-tracking, tight fitting clothing made of special materials.	**Sensors:** Heart rate, breathing rate sensors and step counter. **Connectivity:** Bluetooth. **Materials:** Conductive yarns and quick dry fabric. **Battery Type/Capacity/Life:** -/-/Up to 10 workouts or 1 day when fully charged. **Price:** $69-169
	Citizen Sciences [115]	Fitness-tracking, tight fitting clothing embedded with micro-sensors.	**Sensors:** Heart rate, breathing rate, temperature sensors and step counter. **Materials:** Smart fabrics.
	Physiclo [116]	Compression tights with resistance that improve the training outcomes like burning more calories.	**Sensors:** Surface electromyography (sEMG) electrodes positioned over target muscle groups are used to measure the amount of electrical activity due to muscle contraction. **Price:** $95-125
	MYONTEC [114]	Muscle sensing smart shorts that monitor the wearer′s bio-signals such as muscle load, balance, heart rate, cadence, speed and distance.	**Charging Method:** Charging via USB power cable. **Connectivity:** Bluetooth and USB. **Price:** $918
	Project Jacquard [156]	An industrialised process to integrate elastic electronics into textiles. A currently project is the Levi′s Commute Trucker Jacket. The jacket is designed as a phone extension for navigation, playing music, answering/dismissing phone calls. It has circuits woven on one sleeve with a special button that senses touch/swipes.	
	Ofseth [323]	It is a fiber-optic embedded textile that monitors respiratory movement and cardiac rate.	**Sensors:** A macrobending sensor, a sensor based on fibre Bragg gratings (FBG), and a sensor using the optical time-domain reflectometry (OTDR) in polymer optical fibre (POF).
	Neutralite [324]	The interactive dresses are activated by the spectator′s voice and respond to the spectator′s gaze through the motion of pins which is full of uncertainties.	
	Myovibe [68]	A wearable mechanomygraphy (MMG) system *MyoVibe* was developed to detect the activation of skeletal muscles.	
	Dopplesleep [69]	A contactless sleep monitoring system designed using continuous wave Doppler radar. It detects breathing rate, heart rate and body movements to classify different sleep stages.	

Nevertheless, we do not see much interaction among current wearables primarily due to the significant gap in the software development in both the operating system level and the application level. Therefore, we believe that wearable software will considerably improve in the recent future to harness the advanced hardware capabilities offered by wearables.

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TABLE XIII
E-TEXTILE (CONTINUED)

E-Textile			
Sub Category	**Product**	**Description**	**Technical Specification**
Foot / Hand -worn	ProGlove [325]	A wearable designed for industrial use, i.e. production and logistics by scanning of 1D/2D barcodes.	**Battery Type/Capacity/Life:** -/-/8−10h **Connectivity:** Wireless connection (sub 1 GHz) **Sensor:** Barcode scanner, inertial measurement unit.
	GPS Smartsole [135]	For family members or caregivers to track the wearer′s location through a smartphone, tablet or web browser.	**Battery Type/Capacity/Life:** -/-/Up to 1−2 days. **Price:** $299
	Sensoria [71]	Socks embedded with sensors that count actual steps taken and provide accurate cadence monitoring.	**Connectivity:** Bluetooth. **Sensor:** sensors on the sock count actual steps taken and provide accurate cadence monitoring. **Price:** $69-399
	Owlet Smart Sock [326]	Sock that monitors a baby′s health status in sleep and alerts the parents if the child stops breathing, experiences a drop in blood-oxygen levels, or has irregular vital signs.	**Battery Type/Capacity/Life:** Rechargeable coin-cell battery/-/-. **Connectivity:** Bluetooth, WiFi. **Sensors:** 4 pulse oximeters. **Price:** $299.99
	Sumsang IOFIT [327]	It is designed for real-time coaching of golf and fitness. It provides users with data on step count, duration and distance.	**Charging Method:** Wireless charging, up to 7−10 days. **Sensors:** motion and pressure sensors.
	OpenGo Sensor Insole [328]	A shoe insole designed for the analysis of the plantar pressure distribution, contact forces and dynamics of the human foot.	**Charging Method:** Wireless charging, up to 7−10 days. **Sensors:** Capacitive pressure sensing pads, 3D accelerometers.
	SolePower Ensole [329]	Shoe insoles converting steps into electrical energy.	**Charging Method:** Kinetic charging. **Sensors:** Temperature, motion. **Location Support:** GPS, RFID.
	Lechal [70]	Shoes or soles that measure steps, distance and calories.	**Battery Type/Capacity/Life:** -/-/Up to 15 days. **Charging Method:** USB charging. **Dimensions:** $24.2 * 41.5$ (mm). **Weight:** 18g. **Connectivity:** Bluetooth. **Price:** $179.99
	LookUp [73]	A shoe-based sensing and alerting system consisting of inertial sensors which alerts distracted pedestrians when they cross the road.	
	Gait Analysis Foot Worns [74]	Correlation between several foot clearance parameters and gait speed using shoe-worn inertial sensors.	
	Foot-worn Inertial Sensors [75]	A 2D kinematic model used to calculates the gait spatiotemporal parameters.	

Along with the trend of standalone wearables, in the future it is possible that other devices, e.g., *Smart watches* and *Smart glasses*, may also act as the coordinator and/or the data aggregator within the body, in addition to smartphones. On one hand, this will enable new research directions in distributed system design for wearable networks [304]. For example, multiple wearables will be able to host/store personal data collected from other wearables. On the other hand, multiple contact points to outside networks (e.g., Internet) significantly increases the potential of security attacks, especially when the devices are low-resourced. Thus, distributed secure storage on wearables through emerging technologies such as Blockchains could be another potential research direction.

Management of personal wearables will also become a challenge for users in the future. For instance, recharging all devices, keeping track of battery states of each wearable and making sure device firmware is up to date may not be easy tasks to mange in day-to-day life. It might also break the promise of wearables to provide non-invasive services. Thus, there is a requirement for wearable device management platforms that keep track of everything from battery to security of devices and make some decisions without the user intervention. It is also possible that some wearables may become semi-permanent on the body for the ease of management.

VII. CONCLUSION

Interest in wearables is growing rapidly. In the last few years, a large variety of wearables have hit the market. At the same time, the number of research articles related to wearables has been increasing exponentially. In this article, we have surveyed more than 100 commercially available products and classified them according to their functionalities and wearing modes. Our survey has showed that communication security, energy efficiency, and wearable computing are the most studied topics in the literature. We have surveyed the identified communication security issues for wearables and discussed various approaches to address them followed by a survey of research addressing the energy efficiency issues of wearables. We also surveyed wearable computing research topics such as offloading and in-device machine learning. Finally, we have discussed the potential new directions for wearable products development as well as research.

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TABLE XIV
E-PATCHES

E-Patches			
Sub Category	**Product**	**Description**	**Technical Specification**
Sensor patches	e+ Emergency Patch [138]	Designed for situations where medical attention is not accessible, i.e. during a flight.	**Sensor:** ECG sensor, respiratory band and pulse oximetry sensor.
	Chrono Therapeutics [142]	A patch for smoking cessation. The patch tracks the personal use of medication and deliver a proper amount of nicotine replacement therapy (NRT) to the user before they have cravings.	**Battery Type/Capacity/Life:** Coin-cell battery/-/10 week life time.
	Proteus Health Patch [139]	Besides the regular health tracking capabilities (i.e. steps, activity, heart rate tracking), the patch is also able to track medication-taking of their sensor embedded pills.	**Battery Type/Capacity/Life:** -/-/Lasts for about 7 days.
	HealthPatch MD [76]	It detects vital signs including single−lead ECG, heart rate, heart rate variability, respiratory rate, skin temperature, body posture, fall detection and steps.	**Battery Type/Capacity/Life:** Disposable zinc air battery/-/3 days with ECG, 4 days without ECG. **Dimensions:** $115 * 40 * 7$ (mm) **Weight:** 11 g with Adhesive Patch and Sensor Module. **Sensors:** ECG electrodes, 3-axis MEMS accelerometer, thermistor.
	Thync [77]	A strip that provides relief to the wearer to lower the stress level. It works by generating low-level electric pulses to stimulate nerves on the neck and head which will then affect the brain′s adrenaline release.	**Battery Type/Capacity/Life:** -/-/4-6 vibes per charge. **Charging Method:** USB charging. **Weight:** 18g. Connectivity: BLE.
	Valedo Back Therapy [307]	It keeps track of the development and progress of the patient′s training.	**Charging Method:** with a USB charging cable, run for $9 - 10hr$ with a 45min charge. **Dimensions:** $42 * 32 * 16$ (mm). **Weight:** 17g. **Connectivity:** Bluetooth. **Price:** $359
	OmniPod Insulin Management System [26]	An insulin-delivery system for Type 1 Diabetes.	**Battery Type/Capacity/Life:** 2 AAA alkaline/-/3 weeks. **Dimensions:** $1.6*2.4*0.7$(inches) (Pod), $2.5*4.5*1.0$ (inches) (PDM) **Weight:** 1.2 ounces with full reservoir, 4.4 ounces with batteries. **Connectivity:** Wireless, secure, bidirectional radio frequency.
E-Skin/ E-Tattoos	Motorola e-tattoo [82]	This is applied to the throat area.	**Sensors:** An embedded microphone, a transceiver for enabling wireless communication with the MCD (mobile communication device), a controller coupled to the power supply.
	My UV Patch [143]	A tattoo-like device made of stretchable electronics that detects the amount of UV exposure a wearer receives.	**Connectivity:** NFC.
	Electronic Fingertip [161]	The electronic fingertip utilises flexible electronic technologies. It senses stresses and strain at the finger tip and assess electrical properties of tissue to help doctors remove the tissue precisely by local ablation.	**Sensors:** Pressure and strain sensors. **Material:** Conductive lines and ultra-thin silicon layers.
	Optoelectronic Skin [330]	Blood oxygen concentration level monitoring.	**Energy:** Up to several days. **Sensors:** optical sensors.
	Tattoo-Based Iontophoretic-Biosensing System [80]	An iontophoretic-biosensing-based alcohol sensing device built with flexible wireless circuits. It measures the Blood Alcohol Concentration (BAC) from the sweat and transmits data via bluetooth.	
Others	*Sarotis [331]*	Sarotis is designed under the context of navigation and virtual reality. The soft prosthesis amplifies user's awareness of space using 3D vision technologies.	**Actuators:** Soft amplifiers.
	Smart Tooth Patch [81]	A research prototype designed to sense activities in the mouth like chewing and drinking.	**Sensor:** Accelerometer.
	DuoSkin [79]	A fabrication process of on-skin user interfaces. Using gold leaf as the key material, the products are integrated with functionalities like sensing touch input, thermochromic display and communicating wirelessly using NFC.	

APPENDIX

Table III lists the products and prototypes we surveyed according to their category and applications. Tables IV–X, XII–XIII, and XIV provides further details on these products based on their category. Finally, Table XV and Table XVI compare product specifications of 10 popular smart watches and 10 popular wrist bands.

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TABLE XV
TECHNICAL SPECIFICATIONS OF 10 POPULAR SMART WATCHES
NA* - INFORMATION NOT AVAILABLE

Model	**Apple Watch Series 2** [30]	**Moto 360 (42 mm)** [32]	**Sumsang Gear S2** [31]	**Sony Watch 3** [332]	**Huawei Watch** [86]	**LG urbane 2** [89]	**Fitbit Surge** [37]	**FOSSIL Q FOUNDER** [333]	**ASUS ZenWatch 3** [334]	**Pebble Time 2** [33]
Battery	Lithium-Ion	Lithium-Poly 300 mAh	Lithium-Ion 250 mAh	Lithium-Poly 420 mAh	Lithium-Ion 300 mAh	Lithium-Ion 570 mAh	Lithium-Poly	Lithium-Ion 400 mAh	340 mAh	NA
Continuous HR Monitoring	Yes	Yes	Yes	No HR monitor	Yes	Yes	Yes	No HR monitor	No HR monitor	Yes
CPU	Apple S2 Dual-core	Qualcomm Snapdragon 400 Quad-core CPU (APQ 8026), 1.2 GHz	Exynos 3250, 1GHz Dual Core	Cortex-A7 Quad-core, 1.2 GHz	Cortex-A7 Quad-core, 1.2 GHz	Qualcomm Snapdragon 400 Cortex-A7 Quad-core, 1.2 GHz	32-Bit ARM Cortex-M3, 48MHz	Intel Atom Z34XX CPU	Qualcomm Snapdragon Wear 2100 Quad-core	NA
GPU	NA	Adreno 305, 450MHz GPU	NA	NA	Adreno 305	Adreno 305	NA	NA	NA	NA
Connectivity	BLE 4.0 Wi-Fi 802.11b/g/n 2.4GHz NFC	BLE 4.0 Wi-Fi 802.11 b/g NFC	BT4.1, Wi-Fi, NFC, 3G	BT4.0, Wi-Fi, NFC, Micro USB	BT4.1, Wi-Fi, NFC	Carrier data plan BT 4.1, BLE Wi-Fi 802.11 b/g/n 2.4 GHz	BT 4.0, USB, NFC	BT4.0+, Wi-Fi	BT4.1	BLE 4.0, NFC
GPS	Yes	No	No	Yes	No	No	Yes	No	No	No
Memory		4 GB	4 GB	4 GB	NA	4 GB	1024KB Flash	4 GB	4 GB	
RAM		512 MB	512 MB	512 MB		768 MB	128 KB	1 GB	512 MB	
Display	ION-X strengthened glass 2nd-generation OLED Retina display 326 dpi (272 x 340)	Corning Gorilla Glass 3 Backlit LCD 1.37 in (35mm) 263 ppi (360 x 325)	Super AMOLED, 302 ppi (360 x 360)	TFT capacitive touchscreen 65K colors 283 ppi (320 x 320)	AMOLED capacitive touchscreen 16M colors 286 ppi (400 x 400)	P-OLED capacitive touchscreen 16M colors 348 ppi (480 x 480)	1.2 in Diagonal, Transflective Monochrome with Integral Flex PCB, with 1 LED Backlight 175 ppi (144 x 168)	1.5 in LCD display 240 ppi (360 x 326)	AMOLED capacitive touchscreen 16M colors 287 ppi (400 x 400)	Curved Corning Gorilla Glass (200 x 228)
Speaker/Mic	Yes	Dual digital mics	Yes	Yes	Yes	Yes	No	Yes	Yes	No
Release Date	07/09/2016	14/09/2015	2015	Announced at IFA 2014	02/09/2015	03/2016	NA	NA	11/16	30/06/2016
Price	$349	Starting from $349.99	$250-350	$249.99	$149-349	$359.99	$249.95	$275-315	$299	$169

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TABLE XVI
TECHNICAL SPECIFICATIONS OF 10 POPULAR WRIST BANDS
NA* - INFORMATION NOT AVAILABLE

Model	**UP4 By Jawbone** [36]	**LG Gizmopal 2** [162]	**Fitbit Flex 2** [310]	**Empatica E4 Wristband** [93]	**Nymi Data** [39]	**Microsoft Band** [335]	**Mi Band 2** [309]	**MOOV Now** [38]	**Sony Smart Band 3** [332]	**Huawei Talkband B3** [92]
Battery	Li-Poly 38mAh	Li-Poly 510mAh	Li-Poly	Li-Poly 260mAh	NA	Li-Poly	Li-Poly 41mAh	Coin cell Battery (CR2032)	Li-ion	Li-ion 91 mAh
Continuous Heart Rate Monitoring	No	No	No	Yes	Yes	Yes	Yes	No	Yes	No
CPU	NA	Qualcomm QSC6155	NA	NA	NA	NA	NA	NA	ARM Cortex-M0 32-bit processor	NA
GPU										
Connectivity	BT 4.0	CDMA	BLE	BT Smart	BT4.1, BLE, NFC	BT 4.0, BLE	BT 4.0, BLE	BT	BT 4.0, Wi-Fi, NFC	BT
GPS	No	Yes	No	No	No	Yes	No	No	No	No
Memory (Internal)	NA	NA	NA	NA	NA	NA	NA	NA	NA	512 KB flash
RAM	NA	NA	NA	NA	NA	NA	NA	NA	NA	128 KB
Display	No Display	No Display	No Display	No Display	No Display	AMOLED 32mm x 12.8mm (320 x 128)	OLED display	No Display	No Display	PMOLED touchscreen
Speaker/Mic	No	Yes		No	No	No	Yes	No	Yes	Yes
Release Date	NA	06/11/2015	29/08/2016	NA	04/10/2016	10/2014	07/06/2016		29/09/2015	07/2016
Price (USD)	$199.99	$79.99	$89.95-99.95	$1690	$199	Starting from $199	$23-43	$100	$80	$190-282

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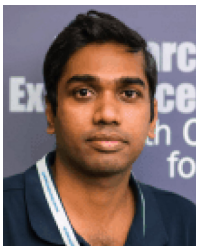
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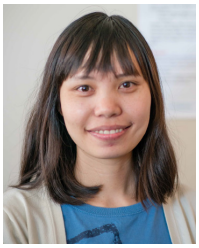
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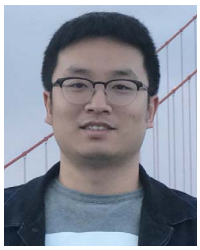
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